



École Polytechnique Fédérale de Lausanne

Forecasting Emerging Technologies: Integrating Scientific and Patent Signals

Master Thesis

by

Dominique Huang

Supervisors:

Prof. Andrea CAVALLARO
EPFL Supervisor

Prof. Guillaume MULLER
EMSE Supervisor

Julian JANG-JACCARD
Industrial Supervisor

August 18, 2026

Abstract

Identifying emerging technologies from scientific and technological data is challenging because emergence has no single quantitative definition and publication- and patent-based signals capture different aspects of technological development. This thesis proposes a temporal forecasting framework that integrates scientific publications from OpenAlex with patent information from the European Patent Office to identify research areas exhibiting signals consistent with future scientific growth and potential technological transition. The framework is evaluated in the quantum information science domain.

Scientific and patent abstracts are processed using a SciBERT-based concept extraction pipeline, followed by dimensionality reduction, HDBSCAN clustering, and temporal cluster stabilization to construct persistent research clusters. Scientific citation relationships are then represented as a temporal knowledge-flow network, while patent-derived information is incorporated as complementary node, edge, and ranking signals. A LightGBM model uses bibliometric, network, temporal, and patent-derived features to forecast three-year growth in cluster in-strength. The resulting forecasts are used in a two-tier ranking framework: Tier 1 identifies candidates for scientific emergence, while Tier 2 prioritizes growing research areas that additionally exhibit patent-related activity.

The framework is evaluated retrospectively by using information available up to 2022 to forecast growth observed in 2025. Of the 10,784 clusters identified at the 2022 forecast origin, 10,599 could be consistently matched to the 14,367 clusters observed in 2025. The full Tier 1 pipeline places all of its top-10 candidates above the 75th percentile of realized global growth, with a mean global percentile of 97.1%. The ranking achieves a Spearman correlation of 0.718 and an NDCG@10 of 0.897 between the predicted ranking and subsequently observed growth.

A patent-ablation experiment shows that publication and citation-network information alone remains sufficient to retrieve many high-growth research ar-

eas: the patent-free top-10 candidates achieve a mean global growth percentile of 93.8%. However, removing patent-derived information reduces the Spearman correlation from 0.718 to 0.399, NDCG@10 from 0.897 to 0.802, and top-10 growth lift from 7.85 to 5.23. These results indicate that, within the evaluated setting, patent information contributes primarily to the prioritization and ranking of promising research areas rather than being necessary for their basic detection.

Finally, the validated pipeline is retrained using observations available up to 2025 to generate prospective candidates for 2028. These outputs are interpreted as model-generated hypotheses rather than established future technological developments and require validation against future observations.

Acknowledgements

I would also like to express my gratitude to Julian Jang-Jaccard for her supervision and support during my work at CYD Campus. The opportunity to collaborate on an industrially relevant problem provided valuable insights into the challenges of technology forecasting and the application of research methods in practice. I also want to thank my supervisor at EPFL Andrea Cavallaro and at EMSE Guillaume Muller for their guidance. Finally, I would like to thank the members of CYD campus for the stimulating research environment and the discussions that contributed to this thesis. I am grateful for the support and encouragement received throughout this journey.

Contents

1 Introduction **7**

2 Related work **11**
2.1 AI-Driven Methods 11
2.2 Research Trend Forecasting 13

3 Methodology **16**
3.1 Dataset Creation 17
3.2 Concept Extraction 20
3.2.1 Clustering 22
3.3 Graph Construction 28
3.3.1 Cumulative Graph and Network Metrics 36
3.4 Forecasting 37
3.4.1 Forecasting Setup 37
3.4.2 Scoring and Ranking 39

4 Validation and Results **46**
4.1 Validation 46
4.1.1 In-Strength versus In-Degree as the Forecasting Target . . . 47
4.1.2 Feature Importance 48
4.1.3 Selection of Patent Boost Coefficients. 49
4.1.4 Held-Out Validation Against 2025 Observations 52
4.1.5 Patent Ablation Analysis 56
4.2 Results 59
4.2.1 Tier 1: Scientific Emergence Forecasting 59
4.2.2 Tier 2: Patent-Informed Technology Transition 60
4.2.3 Qualitative Interpretation of the 2028 Forecast 62

5 Limitations and Future Improvements **63**

5.1 Limitations 63
5.2 Future Improvements 67

6 Conclusion **69**

Chapter 1

Introduction

Emerging and Disruptive Technologies (EDTs) such as 5G, quantum computing, and AI are innovations that are changing the current ecosystem and the national security environment. Despite extensive discussion of EDTs and their growing importance within cybersecurity and national defense contexts [1, 2], technology maturity is predominantly assessed using qualitative metrics such as surveys, expert assessment by research firms such as Gartner, Deloitte, and others. Quantitative methods and metrics can provide a more systematic approach to forecasting emerging and disruptive technologies, and are the center of multiple research efforts, and multiple methods are being created to tackle this problem [31, 32]. However, using quantitative methods mean that we need a data source. Creating a high-quality dataset is essential, as it directly influences the ability to extract meaningful concepts and rank them as emerging technologies.

Gu and Krenn [3] relied on scientific papers collected from arXiv, bioRxiv, ChemRxiv, and medRxiv, while Maillart et al. [4] used OpenAlex to enable large-scale monitoring of scientific evolution due to its extensive coverage. Yang et al. [5] focused on biomedical domains and used PubMed Central to track and predict the emerging topic of research. Norouzi et al. [7] used the Inspec and SemEval-2017 datasets, which contain abstracts of scientific articles with manually annotated concepts, allowing a controlled evaluation of concept extraction and ranking methods. Independently of papers, patent was also used in methods. Daim et al. [8] used Lens.org cybersecurity patents to analyze groupings of topics and identify the leading assignees in this domain.

However, an important challenge in quantitative EDT forecasting concerns the integration of heterogeneous data sources. Many existing quantitative ap-

proaches rely primarily on a single type of signal or data source for technological emergence. Combining different types of data sources can provide a broader perspective and potentially improve the accuracy of EDT forecasting. The complementary relationship between academic literature and patents has been well established in the bibliometric literature. Narin et al. [35] demonstrated that U.S. patents increasingly cite academic publications, providing evidence of a relationship between scientific research and industrial innovation. This relationship motivates the use of patent activity in the present study as a potential complementary signal of technological and industrial relevance, rather than as a definitive leading indicator of future scientific development.

Building on this work, Meyer [36] examined patent–paper citation links in emerging technological fields, highlighting the interaction between academic and technological knowledge production. These findings further motivate the use of patent information as a complementary signal that may capture aspects of technological development not directly reflected in publication and citation activity alone.

Together, these studies provide a strong empirical foundation for incorporating patent information into technology forecasting. Dolamic et al. [6] combined USPTO PatentsView with arXiv and Wikipedia data to identify emerging disruptive technologies and Ghaemmaghami et al. [34] used both papers and patents to forecast emerging technologies in the field of AI. However, these studies provide limited detail on how heterogeneous publication and patent information is integrated into a unified representation. Such integration is non-trivial because scientific and patent abstracts differ in purpose, structure, and terminology, with patent documents typically containing more technical and legal language.

Rather than treating these data sources in isolation, we construct a unified temporal citation network in which scientific citation relationships define the underlying graph, while patent-derived signals are used to reweight its nodes and edges. This approach enables us to evaluate the complementary predictive contributions of scientific activity and patent information through a multi-stage pipeline spanning concept extraction, temporal clustering and stabilization, graph construction, feature engineering, and forecasting.

A fundamental challenge in automated EDT forecasting is the lack of a unified quantitative definition of what constitutes "emergence." In this study, an emerging technology is operationally defined not as a completely unprecedented invention, but as an existing research area exhibiting signals consistent with strong future growth. Three-year growth in cluster in-strength is used as the

primary forecasting target and as the principal prospective signal of scientific emergence, defined as the sum of the weighted incoming citation edges associated with a research cluster. Unlike a simple publication or citation count, weighted in-strength captures the strength of incoming relationships from other areas of the scientific network, with patent-derived signals providing additional weighting in the full representation. Strong future growth in this measure is consequently interpreted as a signal of increasing scientific influence. Patent-derived information provides an additional dimension by capturing evidence of technological and industrial activity. The objective is therefore not to predict the appearance of entirely new scientific concepts, but to identify existing research areas that may be entering a period of rapid scientific and technological development.

Based on these considerations, this thesis addresses the following research questions:

- **RQ1:** Can bibliometric and citation-network signals be used to forecast research clusters exhibiting strong growth in scientific influence over a three-year horizon?
- **RQ2:** Does incorporating patent-derived information improve forecasting and ranking performance compared with publication-only signals?

The main contributions of this thesis are as follows:

- A temporal framework for extracting, clustering, and stabilizing scientific concepts across yearly observations.
- A unified citation-network representation in which scientific citation relationships define the underlying graph and patent-derived signals are used to reweight nodes and edges.
- A two-tier ranking framework distinguishing scientific emergence from patent-informed technological transition.
- A retrospective temporal validation and patent-ablation analysis that evaluates the contribution of patent information to the detection and prioritization of emerging research areas.

We validate this framework through a comprehensive case study in the quantum information science domain. The patent-ablation experiment shows that citation-network and bibliometric signals alone remain capable of identifying many research areas that subsequently experience strong growth. However, integrating industrial patent signals substantially improves how accurately these candidates are prioritized, increasing the Spearman rank correlation from 0.399 to 0.718. This indicates that patent information serves primarily as a refinement

mechanism rather than as a prerequisite for detecting them in the first place.

The remainder of this thesis is structured as follows. Section 2 reviews existing quantitative and AI-driven approaches for identifying and forecasting emerging technologies. Section 3 presents the proposed methodology, including data collection, concept extraction, temporal graph construction, patent integration, and forecasting. Section 4 presents the experimental results and validation. Section 5 discusses the limitations of the proposed approach and potential improvements. Finally, Section 6 summarizes the main conclusions and outlines directions for future research.

Chapter 2

Related work

2.1 AI-Driven Methods

With an ever increasing number of data sources, it is important to use meaningful techniques to treat all those data. Deep learning thus became a powerful method to leverage knowledge from large and sparse data sources. Moreover, those methods are also used in the forecasting part.

Most of the time, the first step of forecasting an EDTs is extracting keywords and concepts from different abstracts or titles; we want this extraction to be accurate and precise to have a better forecasting. Those extracted concepts can also be used in others following methods to have better forecasting, such as creating a graph using those concepts and thus showing the evolving landscape of science and the identification of EDTs. Various concepts extracting methods exist whether using Large Language Models (LLM) or not. Traditional keyword extraction is mostly frequency based, such as RAKE (Rapid Automatic Keyword Extraction) [11], TF-IDF (Term Frequency-Inverse Document Frequency) [12] or YAKE (Yet Another Keyword Extraction) [13]. Those techniques are statistical and cannot use semantic information from data. Another type of keyword extraction techniques are graph based, such as TextRank [15] and SingleRank [16]. In graph based techniques, we represent the document as a graph where the nodes are represented by the keyword and the connection between them are represented by the edges. Then, we use graph based metrics to rank the different keyword and select the top among them.

With recent advancements in language representation and information extraction, transformer-based models such as BERT (Bidirectional Encoder Representations from Transformers) [10] have introduced contextual embeddings that

capture semantic information from text. These embeddings can be combined with similarity measures, such as cosine similarity, to identify semantically relevant terms and concepts. Studies have shown that BERT-based approaches can outperform traditional frequency-based and graph-based keyword extraction methods [14]. Based on BERT representations, several approaches have been proposed, such as KeyBERT [17], which extracts keywords by comparing candidate phrases with the semantic representation of the document, SciBert [18], which is a BERT model trained on scientific papers, PatentBert [19] trained on patent or SPECTER [20] which learns document-level embeddings for scientific literature.

Beyond concept extraction, clustering methods play a central role in organizing extracted concepts into coherent research topics. Traditional approaches such as K-means [37] require the number of clusters to be specified in advance, which is impractical when the structure of the concept space is unknown. Density-based methods such as DBSCAN (Density-based spatial clustering of applications with noise) [38] overcome this limitation by identifying clusters as dense regions separated by sparse areas, without requiring a predefined cluster count. HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) [28] extends this approach by operating across multiple density scales, making it more robust to clusters of varying density. BERTopic [39] combines transformer-based embeddings with HDBSCAN and a class-based TF-IDF weighting scheme to produce interpretable topic representations. Egger and Yu [40] compared different topic modeling schemes including BERTopic and showed that this model outperforms other models such as LDA (latent Dirichlet allocation) [41] or Top2Vec [42]. In this work, we adopt a similar embedding-plus-clustering strategy using SciBERT and HDBSCAN but operate at the concept level rather than the document level, preserving finer-grained semantic distinctions necessary for technology foresight at the sub-field level.

Once concepts have been extracted and clustered, the resulting representations can serve as input to downstream forecasting models. The use of AI-driven methods extends beyond concept extraction to the forecasting stage itself. Recent studies have applied GNNs (Graph Neural Networks) to citation, co-authorship, and heterogeneous scientific knowledge graphs to predict future research trends and identify emerging technologies [43]. By learning directly from graph structure and node features, GNNs can capture complex relationships that are difficult to model using traditional graph metrics alone. However, these end-to-end approaches generally require large amounts of labeled training

data, are computationally demanding, and often provide limited interpretability.

In contrast, the methodology proposed in this work decouples graph construction from forecasting. The citation graph is first constructed and enriched with semantic and patent-derived information, after which temporal graph features are extracted and used to train a LightGBM model [44]. Gradient boosting methods such as LightGBM are particularly well suited to structured tabular data and offer efficient training, interpretable feature importance, and robust performance with the relatively limited number of yearly observations available in our forecasting setting. This modular design also facilitates the integration of additional data sources or graph features without requiring the forecasting model to be retrained end-to-end.

Overall, in scientometrics, AI-driven methods have become a key approach for understanding the evolution of scientific knowledge and forecasting emerging technologies, offering significant improvements over traditional statistical techniques.

2.2 Research Trend Forecasting

Although AI-driven methods provide powerful tools for extracting and representing scientific knowledge, forecasting the future evolution of research trajectories requires dedicated temporal modeling approaches. Therefore, the following section reviews existing approaches for research trend forecasting and emerging technology prediction.

Traditional approaches generally rely on statistical metrics; for instance, Dolamic et al. [21] computed several indicators, including novelty score, impact score, and growth score. Similarly, Xu et al. [22] used handcrafted indicators, such as the fast growth indicator, radical novelty indicator, and prominent impact indicator, to forecast EDTs. However, recent approaches have shifted toward deep learning-based methods. Gu and Krenn [23] constructed a knowledge graph from research papers and applied machine learning techniques to predict the evolution dynamics of the research landscape with high accuracy. Krenn et al. [24] further leveraged machine learning-based link prediction on a knowledge graph to model the future evolution of AI research topics. Zhou et al. [25] trained a deep neural network classifier capable of predicting up to 77% of emerging technologies in a given year with high precision.

Beyond these indicator-based approaches, weak signal analysis has been

combined with deep learning techniques to identify early signs of technological emergence. Ebadi et al. [33] proposed a framework for detecting emerging technologies by integrating weak signal analysis with deep learning models. Their approach extracts latent patterns from scientific and technological data to identify early-stage technologies before they become widely established. This demonstrates the potential of deep representation learning to capture subtle evolutionary patterns that may not be detectable using traditional bibliometric indicators.

Ghaemmaghami et al. [34] proposed an indicator-based framework that jointly exploits scientific publications and patent data to identify emerging technologies in the field of artificial intelligence. Their approach combines indicators related to growth, novelty, and collaboration to capture both scientific development and technological innovation. By integrating publications and patents, the framework provides a more comprehensive view of technology emergence, linking advances in academic research with their translation into technological applications.

Taken together, these studies demonstrate that technological emergence is generally operationalized through multiple dimensions rather than through a single universally accepted metric. Existing approaches consider indicators such as growth, novelty, impact, collaboration, and weak signals, while other studies exploit citation or graph-based representations to capture the evolution of scientific fields.

The framework proposed in this thesis follows this multidimensional perspective but separates forecasting from the subsequent assessment of emergence. First, three-year growth in cluster in-strength is used as the forecasting target. Rather than measuring growth solely through publication frequency, this target captures the expected change in the strength of incoming relationships received by a research cluster within the scientific citation network. Predicted in-strength growth therefore represents the primary prospective signal of increasing scientific influence, rather than a complete emergence score by itself.

The predicted growth is subsequently incorporated into two ranking levels. Tier 1 combines the forecast with complementary bibliometric, citation, and recent growth indicators to rank general emergence candidates. Tier 2 further introduces explicit patent-activity and patent-momentum criteria to prioritize candidates exhibiting signals consistent with potential technological transition. Patent information is therefore treated as a complementary signal whose contribution is evaluated empirically, rather than as an assumed leading indicator of

technological emergence.

Despite these advances, several limitations remain. Existing indicator-based approaches rely heavily on handcrafted metrics, while many graph-based forecasting methods focus exclusively on scientific publications. The joint exploitation of publications and patents remains relatively underexplored, despite their complementary perspectives on scientific discovery and technological innovation. Furthermore, the semantic differences between scientific articles and patent documents pose additional challenges when constructing unified representations across both domains. These limitations motivate the framework proposed in this thesis, which integrates publications and patents into a unified graph-based forecasting approach while explicitly addressing semantic heterogeneity through embedding-based clustering and similarity techniques.

Chapter 3

Methodology

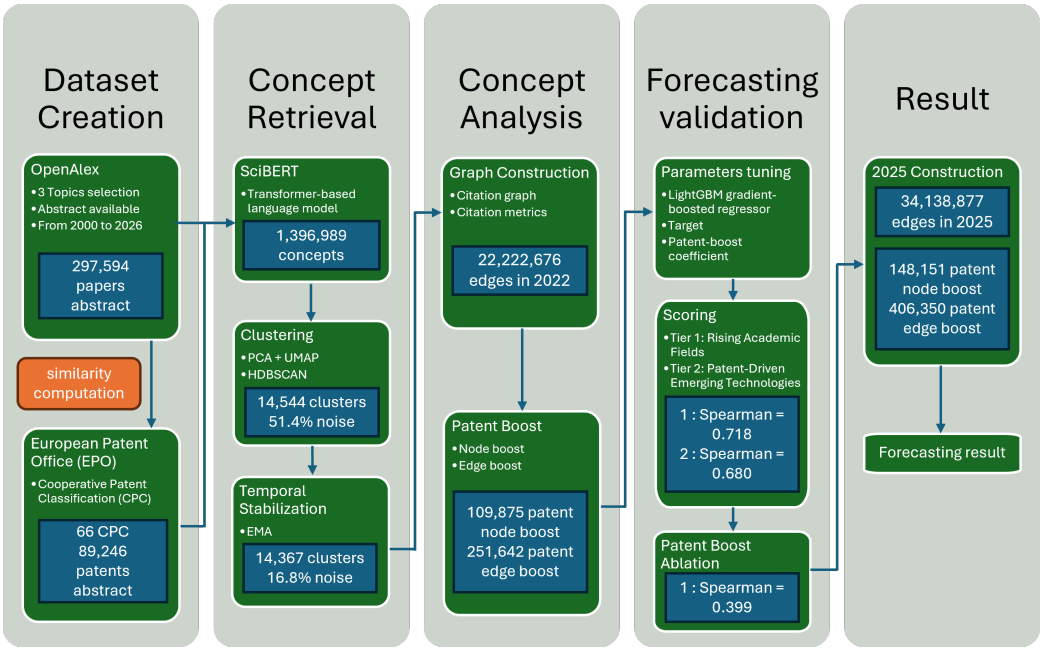


Figure 3.1: Overview of the methodology

Fig. 3.1 presents an overview of the workflow used to process scientific papers and patents for forecasting emerging technologies. The methodology is divided into five main stages: dataset creation, concept retrieval, concept analysis, forecasting validation, and results.

First, we select papers from OpenAlex using predefined topics, restricting the dataset to publications with available abstracts from 2000 to 2026. This results

in a collection of 297,594 papers with abstracts. We then compute similarities between these papers and patent classifiers to retrieve patents whose abstracts belong to the most relevant technological categories.

Next, SciBERT is applied to the retrieved abstracts to extract scientific concepts. These concepts are subsequently clustered and stabilized across years to ensure temporal consistency. This process results in 14,367 concept clusters, with 16.8% of extracted concepts classified as noise, corresponding to concepts that do not belong to any stable cluster.

The resulting clusters are then analyzed using citation graphs constructed from scientific papers, while patent information is incorporated as an additional signal by enhancing both nodes (clusters) and edges (relationships between clusters).

We validate the proposed methodology and optimize selected parameters to improve forecasting performance. This process leads to the definition of two emergence scores: Tier 1, which identifies rapidly growing academic fields, and Tier 2, which captures patent-informed technology transition.

Finally, we perform technology forecasting using data available up to 2025.

3.1 Dataset Creation

The main data source for emerging technology detection remains scientific publications, which can be extracted from different platforms. In this project, we rely on OpenAlex [9] as our primary source of scientific literature. However, research papers mainly represent an early stage of technological development. Therefore, incorporating additional data sources is valuable to capture later stages of the innovation process. One of the most relevant complementary sources is patents. While scientific papers mainly describe ongoing research activities and new ideas under development, patents generally represent technologies that have reached a more application-oriented stage, where organizations seek legal protection and potential commercialization. Consequently, a technology appearing only in scientific literature may indicate an emerging research direction, whereas a technology appearing in both scientific publications and patents may provide complementary evidence of application-oriented or industrial activity. Patent presence is therefore treated in this thesis as a potential signal of technological relevance rather than as evidence that practical adoption or commercialization has occurred. Similar to scientific literature, patents can be retrieved from differ-

ent databases. In this project, we use the European Patent Office (EPO), which provides an API allowing automated access to patent metadata and content.

As a starting point, we focus on scientific publications extracted from OpenAlex. OpenAlex provides several ways to categorize papers, including fields, domains, keywords, and topics. We choose topics as the main categorization mechanism and select three topics strongly related to quantum computing: "Quantum Information and Cryptography", "Quantum Computing Algorithms and Architecture", and "Quantum Many-Body Systems". This selection results in 297,594 scientific papers published since 2000 containing abstracts. In addition to abstracts, OpenAlex provides additional metadata that we retrieve, including citation relationships and the Field-Weighted Citation Impact (FWCI), which measures the relative citation impact of a publication compared to similar publications.

Patent data were retrieved from the European Patent Office (EPO). Rather than manually selecting patent categories, relevant Cooperative Patent Classification (CPC) codes were identified automatically by comparing OpenAlex publication abstracts with CPC descriptions. The search was restricted to the G, H, and Y CPC sections and to hierarchy levels 0–4 to maintain a manageable and sufficiently broad classification space. Table 3.1 shows an example of CPC code.

Level	**CPC Code**	**Description**
0	G	Physics
1	G06	Computing or Calculating; Counting
2	G06N	Computing arrangements based on specific computational models
3	G06N 10/00	Quantum computing, i.e., information processing based on quantum-mechanical phenomena
4	G06N 10/20	Models of quantum computing, e.g., quantum circuits or universal quantum computers

Table 3.1: Breakdown of the CPC classification hierarchy

Relevant CPC codes were identified using a hybrid similarity approach combining lexical and semantic matching. TF–IDF cosine similarity captures direct technical vocabulary overlap between publication abstracts and CPC descriptions, while Sentence Transformer embeddings (`all-MiniLM-L6-v2`) capture semantic similarity when different terminology is used. The two similarity measures were combined into a weighted score, and CPC codes exceeding the

selected similarity threshold were retained for patent retrieval.

Rank	CPC Code	Papers	Avg. Score	Label
1	G06N10/20	6291	0.490	Models of quantum computing, e.g. quantum circuits or universal quantum computers
2	G06N10/60	5351	0.484	Quantum algorithms, e.g. based on quantum optimisation, quantum Fourier or Hadamard transforms
3	G06N10/80	3233	0.490	Quantum programming, interfaces, languages, software-development kits, and quantum computing platforms
4	G06N10/40	2493	0.476	Physical realisations or architectures of quantum processors and qubit manipulation components
5	G06N10/70	739	0.475	Quantum error correction, detection or prevention, e.g. surface codes or magic state distillation
6	G06N10/00	594	0.474	Quantum computing, i.e. information processing based on quantum-mechanical phenomena
7	H04L9/0852	415	0.481	Quantum cryptography
8	H10D48/3835	239	0.477	Semiconductor qubit devices comprising interacting semiconductor quantum dots
9	H04B10/70	82	0.471	Photonic quantum communication
10	H04J14/07	55	0.488	Orbital angular momentum multiplex systems

Table 3.2: Top 10 CPC categories in quantum computing research ranked by number of papers

Table 3.2 presents the ten CPC categories with the highest similarity to the publication abstracts in the dataset. For each CPC code, the table reports the number of papers assigned to that category together with the average similarity score between the paper abstracts and the CPC description. The results show that the dataset is dominated by core quantum computing topics, including quantum computing models, quantum algorithms, quantum programming, quantum processor architectures, and quantum error correction. More specialised areas, such as quantum cryptography, semiconductor qubit devices, and photonic quantum communication, are also represented, although with a

smaller number of publications. Overall, these results confirm that the collected corpus is strongly focused on quantum computing while covering a broad range of its major research directions.

We then utilize those CPC code and retrieve all patents with their abstract from EPO using the API and finally, we go through a cleaning process where we remove the duplicate data. We do not need to clean papers data as in the way that we use the API, the same papers are already regrouped together. We end up with a total of 66 different CPC and a total of 89,246 patents. Here is the evolution of publications over the years in Figure 3.2.

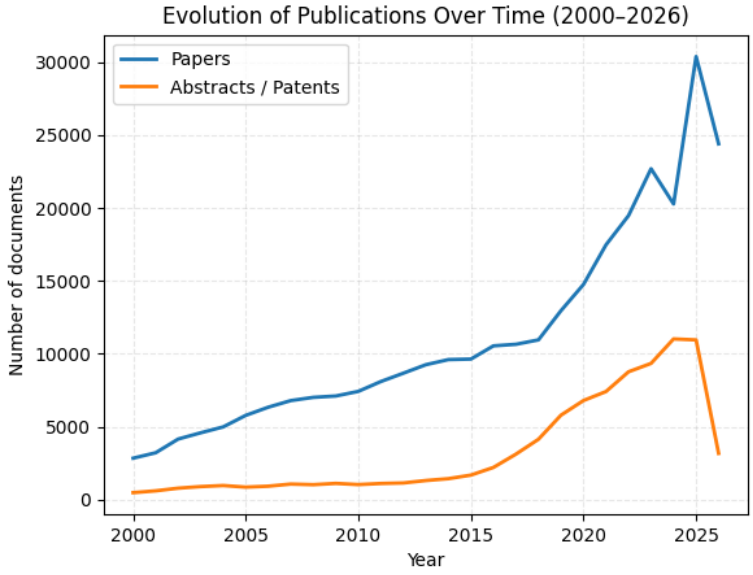


Figure 3.2: Evolution of Publications Over Time (2000–2026)

3.2 Concept Extraction

Once we have our dataset of paper and patent abstracts, we retrieve relevant concepts from these texts. To extract concepts from scientific abstracts, we use a KeyBERT-based approach with SciBERT embeddings. KeyBERT [17] is a keyword extraction method that identifies the most representative terms by measuring the semantic similarity between candidate concepts and the document representation. Instead of relying only on word frequency, we use SciBERT [18], a transformer-based language model trained on scientific literature, to generate

contextual embeddings that capture domain-specific semantic relationships. This allows the extraction process to consider the meaning and context of scientific concepts, which is particularly relevant for papers and patents.

Although SciBERT was trained primarily on scientific publications, we apply it uniformly to both paper and patent abstracts in order to maintain a shared embedding space across both data sources. This design choice places concepts extracted from papers and patents within a common embedding space, allowing their semantic similarities to be evaluated consistently in the downstream clustering and graph construction steps.

This decision is motivated by the substantial linguistic differences between the two domains. Patent abstracts tend to be legally oriented and intentionally broad, describing technologies in less explicit terms in order to maximize coverage and protect intellectual property. As a result, a domain-specific model such as PatentBERT would likely produce higher-quality representations for patents in isolation. However, it would also introduce a domain mismatch, since paper and patent concepts would be embedded in separate vector spaces that are not directly comparable.

Bridging two independently trained embedding spaces typically requires domain adaptation or cross-domain alignment techniques. A large class of supervised alignment approaches relies on shared anchor concepts or paired examples to learn an explicit mapping between embedding spaces. In contrast, unsupervised methods relax this requirement but instead depend on stronger distributional similarity assumptions between the two spaces.

In our setting, only a few concepts exhibit non-zero frequency in both papers and patents for a given year representing less than 1% of the total concept vocabulary. This extremely limited overlap provides insufficient shared structure for anchor-based alignment methods and weakens the assumptions required by unsupervised distribution-matching approaches, making cross-domain alignment unreliable in practice.

Therefore, we prioritize embedding consistency over domain specialization and adopt SciBERT as a unified encoder for both sources. Future work may explore methods for better reconciling the two vocabularies within a shared representation framework.

During concept extraction, we preserve provenance information, i.e., the association between each extracted concept and its source document (paper or patent). This information is later exploited in the graph construction phase, where it removes the need to explicitly match concepts across documents.

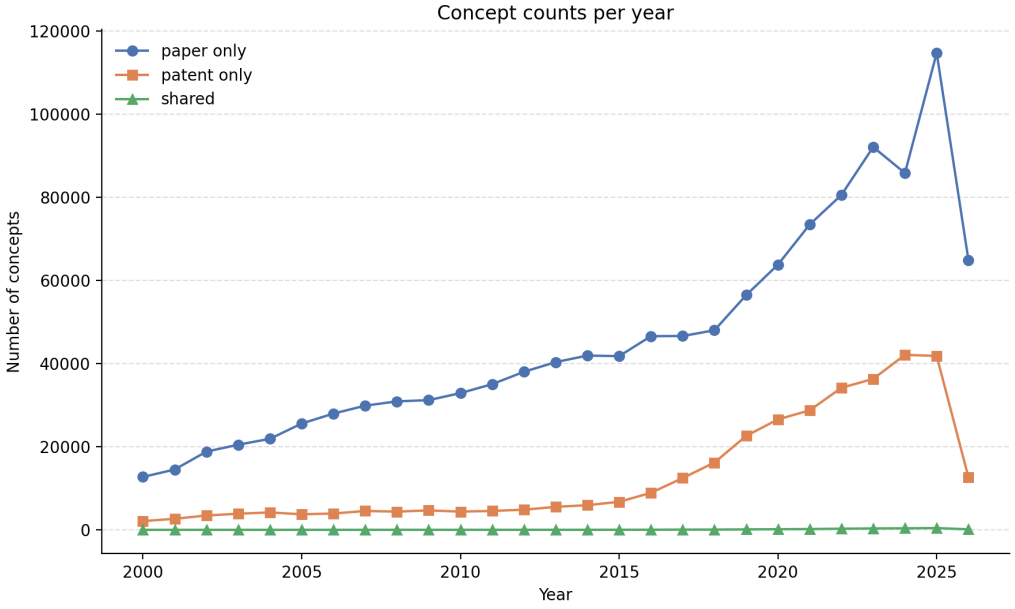


Figure 3.3: Annual evolution of the number of extracted concepts from scientific publications and patents

Figure 3.3 illustrates the evolution of the number of concepts extracted yearly over time. For each document, we retain the five highest-ranked candidate concepts to limit noise. Candidate keyphrases are restricted to two to four words with four-word concepts accounting for 77.57% of the unique vocabulary.

Table 3.3 summarizes the vocabulary produced by the concept-extraction pipeline. Across the 2000-2026 observation period, 1,396,989 unique normalized concepts were extracted. Direct lexical overlap between the publication and patent vocabularies is limited: only 4,450 concepts occur in both sources, corresponding to approximately 0.32% of the combined vocabulary. Nevertheless, several domain-relevant concepts occur frequently in both sources, as illustrated in Table 3.4.

3.2.1 Clustering

The next step consists of clustering extracted concepts in order to address redundancy arising from surface-level variations. After extraction, semantically identical or highly similar concepts may appear under different textual forms.

Table 3.3: Descriptive statistics of the extracted concept vocabulary (2000–2026).

Statistic	**Value**
Observation period	2000–2026
Source documents	386,840
Total concept occurrences	1,921,026
Unique normalized concepts	1,396,989
Unique paper concepts	1,059,108
Unique patent concepts	342,331
Paper-only concepts	1,054,658
Patent-only concepts	337,881
Shared concepts	4,450
Shared vocabulary	0.32%
Mean extraction score	0.701
Median extraction score	0.700

Table 3.4: Examples of concepts occurring in both scientific publications and patents.

Concept	**Paper Occ.**	**Patent Occ.**
Quantum key distribution	1,615	141
Quantum computing	1,452	77
Quantum machine learning	676	17
Quantum random number generator	300	155
Quantum cryptography	421	19
Quantum error correction	408	17
Quantum processor	313	33
Quantum secret sharing	345	29

For example, “quantum key distribution” and “quantum key distribution QKD” refer to the same underlying concept but are represented as distinct strings.

To mitigate this issue, we group such variants into unified clusters. For each cluster, we aggregate metadata such as frequency across documents and occurrences. A representative label is then assigned to each cluster, corresponding to the most frequent or most central concept within the group.

We apply a two-step dimensionality reduction process before clustering. First, PCA (Principal Component Analysis) [26] is used to reduce noise and computational complexity while preserving the most informative variance in the embedding space, we lower the number of dimension of each embeddings and keep only the most important one. This is done to speed up the compu-

tation but also remove low variance dimension which can be considered as noise and thus improves the stability. Second, UMAP (Uniform Manifold Approximation and Projection) [27] is applied to project the embeddings into a lower-dimensional manifold that preserves local semantic structure, which is particularly suitable for downstream density-based clustering. Although both PCA and UMAP perform dimensionality reduction, they serve complementary purposes in the proposed pipeline. PCA first provides a computationally efficient linear reduction that removes low-variance dimensions, whereas UMAP subsequently performs a nonlinear projection designed to preserve local neighborhood structure. Applying PCA before UMAP therefore reduces computational cost while providing a more compact representation for subsequent manifold learning and density-based clustering.

Finally, we use HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) [28] to identify clusters of semantically similar concepts. HDBSCAN is a density-based clustering algorithm that can detect clusters of arbitrary shape and explicitly handle outliers, which are treated as noise rather than forced into clusters. We lower dimensionality before HDBSCAN mainly because HDBSCAN relies on meaningful distance and density estimates, and those become unreliable or expensive in very high-dimensional spaces. The framework is similar to the one that we can find in BERTopic [39].

The HDBSCAN clustering algorithm was configured with `min_cluster_size` = 3 and `min_samples` = 2. Setting `min_cluster_size` to 3 favours the identification of fine-grained semantic clusters while ensuring that emerging research topics represented by only a small number of concepts are retained rather than merged into larger clusters or discarded. The `min_samples` parameter was set to 2, below the default value of `min_cluster_size`. This relaxes the density requirement for a point to be considered a core point, reducing the number of concepts classified as noise. Such a configuration is appropriate for scientific concept embeddings, where newly emerging or highly specialised research topics often occupy sparse regions of the embedding space because of their limited publication volume. Retaining these concepts improves coverage while preserving HDBSCAN's ability to identify dense, well-defined semantic clusters.

The `cluster_selection_method` was set to `leaf` rather than the default Excess of Mass (EoM) criterion. EoM selects clusters by maximizing the total stability of the hierarchy, which tends to produce a small number of large, coarse clusters in practice collapsing distinct sub-fields such as *quantum error correction* and *quantum key distribution* into a single broad *quantum physics* cluster.

The `leaf` method instead selects the most granular partition at the bottom of the condensed tree, preserving the fine-grained topical distinctions necessary for technology foresight at the sub-field level.

Table 3.5: Evolution of HDBSCAN clustering statistics over time.

Year	Concepts	Clusters	Noise (%)	Avg. cluster size
2000	14,853	1,655	39.1	5.46
2005	29,356	3,086	43.9	5.34
2010	37,326	3,886	44.1	5.37
2015	48,597	4,945	46.3	5.28
2020	90,566	8,625	49.7	5.28
2025	157,048	14,544	51.4	5.24

The clustering statistics in table 3.5 show that the selected HDBSCAN configuration produces a highly granular concept partition. The number of identified clusters increases with the size of the concept space, while the average cluster size remains stable around five concepts throughout the observation period. This confirms that the leaf cluster selection strategy preserves fine-grained topical structures rather than producing a limited number of broad research areas.

The proportion of concepts classified as noise increases over time, reaching approximately 50% in recent years. This behavior is expected in a large-scale scientific concept space, where many extracted concepts correspond to specialized or emerging terminology with limited semantic density. Since the objective is early identification of emerging technological trajectories, preserving these fine-grained structures is preferred over aggressively merging concepts into larger clusters.

Since clustering is performed independently for each year, the cluster identifiers produced by HDBSCAN do not have temporal consistency. A cluster assigned ID 5 in one year does not necessarily represent the same concept as cluster ID 5 in another year. Therefore, a temporal alignment step is required to associate clusters across years and create persistent concept identities. There are different methods to align them through the years such as using Jaccard score or containment. However, both those techniques suffer from limitations in this context. The Jaccard similarity between two clusters A and B is defined as:

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

where $|A \cap B|$ represents the number of shared concepts between the two

clusters and $|A \cup B|$ represents the total number of unique concepts. A higher value indicates a stronger overlap between clusters. If for a specific year, we have a cluster with {A,B,C} concepts and then the year after, we have a cluster {A,B,C,D,E,F,G,H,I,J}, ideally, these should be assigned the same cluster ID; however, the Jaccard score would be artificially low due to the fact that the new cluster is bigger. This is very common in this study as when a concept explode and gains a lot of popularity, from year to year, the cluster size might differ a lot. Containment is an overlap-based cluster evolution metric. Unlike Jaccard similarity, it is directional: it asks whether an old cluster is preserved inside a new cluster and is represented by :

$$C(A,B) = \frac{|A \cap B|}{|A|}$$

where A represents the cluster from the previous year, B represents the cluster from the current year, and $|A \cap B|$ denotes the number of shared concepts between the two clusters. This also does not work because it compares exact wording to compute the intersection which can be low because the same concept might have a different wording.

We introduce a cluster alignment and tracking mechanism based on cosine similarity matching of embedding centroids, with Exponential Moving Average (EMA) updates [29] to allow cluster representations to evolve gradually across years.

Each cluster C_i^t is represented by a semantic centroid computed as the mean of its concept embeddings:

$$\mathbf{c}_i^t = \frac{1}{|C_i^t|} \sum_{x \in C_i^t} \mathbf{e}_x, \quad \text{with } \|\mathbf{c}_i^t\|_2 = 1,$$

where $\mathbf{e}_x$ denotes the normalized SciBERT embedding of concept x.

To associate clusters across years, we compute cosine similarity between the centroid of a current cluster and all previously stored stable cluster centroids:

$$\text{sim}(\mathbf{c}_i^t, \mathbf{s}_k) = \mathbf{c}_i^t \cdot \mathbf{s}_k,$$

where both vectors are L_2-normalized. A match is accepted if the maximum similarity exceeds a predefined threshold:

$$\max_k \text{sim}(\mathbf{c}_i^t, \mathbf{s}_k) \geq \tau, \quad \tau = 0.85.$$

If a match is found, the cluster is assigned the corresponding stable identifier. Otherwise, a new stable cluster is created.

To account for gradual semantic evolution, we update matched cluster representations using an EMA:

$$\mathbf{s}_k \leftarrow (1-\alpha)\mathbf{s}_k + \alpha \mathbf{c}_i^t, \quad \alpha = 0.3,$$

followed by re-normalization:

$$\mathbf{s}_k \leftarrow \frac{\mathbf{s}_k}{\|\mathbf{s}_k\|_2}.$$

This mechanism ensures that cluster representations evolve smoothly over time while preserving historical semantic information.

Finally, noise points (HDBSCAN outliers) are assigned to the nearest stable cluster if their cosine similarity exceeds a predefined threshold, subject to capacity constraints proportional to the cluster size. This step further improves temporal consistency by ensuring that isolated or weakly connected concepts are not discarded but instead integrated into semantically coherent clusters when appropriate.

This mechanism enables the harmonization of cluster identities over time. For instance, a concept such as *quantum key distribution* may be assigned to cluster 5 in one year and cluster 13 in the following year due to independent clustering. After applying the temporal alignment procedure, both occurrences are consistently mapped to the same stable cluster identifier.

Moreover, this noise assignment step is particularly important for linking concepts extracted from both patents and scientific papers. As previously discussed, patent abstracts often exhibit a more heterogeneous and legally oriented vocabulary, which leads to a higher proportion of isolated concepts that are not initially assigned to any cluster by HDBSCAN. These concepts are therefore frequently labeled as noise or singletons.

Table 3.6 shows that HDBSCAN alone assigns between 48.6% and 60.9% of extracted concepts to clusters, with the remaining concepts classified as density-based noise. After temporal alignment and semantic reassignment, the final coverage increases to 83.2–90.0% of concepts associated with persistent technological trajectories.

This demonstrates that the proposed stabilization procedure substantially improves temporal continuity while preserving fine-grained semantic structures.

Table 3.6: Evolution of concept coverage before and after temporal stabilization.

Year	Concepts	Stable clusters	HDBSCAN coverage (%)	Stable coverage (%)	Avg. stable size
2000	14,853	1,645	60.9	90.0	8.13
2005	29,356	3,090	56.1	88.4	8.40
2010	37,326	3,882	55.9	89.1	8.56
2015	48,597	4,963	53.8	87.7	8.59
2020	90,566	8,586	50.3	84.5	8.91
2025	157,048	14,367	48.6	83.2	9.09

The remaining unassigned concepts correspond mainly to highly sparse or weakly represented concepts for which no sufficiently similar trajectory exists.

Algorithm 1 and 2 summarizes the temporal cluster stabilization procedure used to assign persistent identifiers to concept clusters across consecutive years. Since HDBSCAN performs clustering independently for each yearly dataset, the resulting cluster labels are not consistent over time and therefore cannot be used directly for longitudinal analysis. The proposed algorithm addresses this limitation by representing each cluster with the centroid of its SciBERT concept embeddings and matching these centroids to those of previously observed clusters using cosine similarity. Matched clusters retain their existing stable identifier, while unmatched clusters are assigned a new one. Finally, concepts initially classified as noise by HDBSCAN are compared with the stable cluster centroids and reassigned whenever their semantic similarity exceeds a predefined threshold. This procedure enables the construction of temporally consistent concept trajectories while substantially increasing the proportion of concepts associated with stable technological clusters.

3.3 Graph Construction

There are different strategies to construct a graph. A co-occurrence graph links concepts that appear within the same document, capturing local semantic proximity inside an article or patent. In contrast, a citation-based graph leverages inter-document relationships: when one paper cites another, we connect the concepts in the citing paper to the concepts in the cited papers. This approach is particularly well-suited for scientific publications because citations in academic literature are generally intended to reflect intellectual influence and knowledge flow between research works. Using OpenAlex, we can retrieve the

Algorithm 1 Temporal Stable-ID Assignment

Require: Clustered concepts D_y for year y, global centroid store $\mathscr{C}$, next available stable ID n

Ensure: Stable IDs for all non-noise clusters and updated centroid store $\mathscr{C}$

Parameters: $\tau_{\text{match}} = 0.85$, $\alpha = 0.3$

Embed all unique concepts in D_y using SciBERT: $\phi(w) \in \mathbb{R}^{768}$
for all local clusters $k \neq -1$ in D_y **do**
    Compute centroid $\mathbf{c}_k \leftarrow |k|^{-1} \sum_{w \in k} \phi(w)$
    Normalize $\mathbf{c}_k \leftarrow \mathbf{c}_k / \|\mathbf{c}_k\|_2$
end for
if $\mathscr{C}$ is empty **then**
    for all local clusters k **do**
        Assign stable ID n to k
        $\mathscr{C}[n] \leftarrow \mathbf{c}_k$
        $n \leftarrow n + 1$
    end for
else
    Compute cosine similarities $S_{ij} \leftarrow \mathbf{c}_i^\top \mathscr{C}[j]$
    $used \leftarrow \emptyset$
    for all local clusters i **do**
        $j^* \leftarrow \arg\max_{j \notin used} S_{ij}$
        if $S_{ij^*} \geq \tau_{\text{match}}$ **then**
            Assign stable ID j^* to i
            $\mathbf{u} \leftarrow (1-\alpha)\mathscr{C}[j^*] + \alpha \mathbf{c}_i$
            $\mathscr{C}[j^*] \leftarrow \mathbf{u} / \|\mathbf{u}\|_2$
            $used \leftarrow used \cup \{j^*\}$
        else
            Assign new stable ID n to i
            $\mathscr{C}[n] \leftarrow \mathbf{c}_i$
            $used \leftarrow used \cup \{n\}$
            $n \leftarrow n + 1$
        end if
    end for
end if
return updated D_y, $\mathscr{C}$, and n

Algorithm 2 Noise Concept Reassignment

Require: Yearly concepts D_y after stable-ID assignment, centroid store $\mathscr{C}$
Ensure: Reassigned noise concepts and updated stable-cluster labels
Parameter: $\tau_{\text{noise}} = 0.78$

for all concepts w with label -1 in D_y **do**
  $j^* \leftarrow \arg\max_j \phi(w)^\top \mathscr{C}[j]$
  $s^* \leftarrow \phi(w)^\top \mathscr{C}[j^*]$
  if $s^* \geq \tau_{\text{noise}}$ **and** noise_count$[j^*] < 0.5\,|\text{cluster}_{j^*}|$ **then**
    Assign w to stable cluster j^*
    noise_count$[j^*] \leftarrow$ noise_count$[j^*] + 1$
  else
    Keep w as noise (stable_cluster $= -1$)
  end if
end for
Save updated $\mathscr{C}$ and n
return D_y with stable-cluster assignments

"referenced_works" information of a paper, which allows us to easily reconstruct the citation network. This makes the resulting graph more structured and semantically meaningful at the level of scientific discourse, as it captures how ideas are built upon and reused over time rather than only co-occurring within a single text. In contrast, patent concepts are considerably sparser than paper concepts and exhibit low overlap with the paper citation network, making it impractical to construct a separate patent graph and subsequently align it with the paper graph. Instead, we integrate patent signal directly into the paper-based graph through a boost mechanism applied to both nodes and edges. The resulting graph is therefore a directed citation graph over concept clusters, enriched by patent co-occurrence signal. The concept will contribute to the cluster node that it belongs to. Therefore, we use data at concept level and the result will be integrated at cluster level to create the graph.

Concept-to-cluster mapping. Each concept x extracted from a paper or patent is mapped to a stable cluster $\phi(x) \in \mathscr{C}$ via the global concept-to-cluster map built in the previous step. Each document d is thus associated with a set of clusters:

$$\Phi(d) = \{\phi(x) \mid x \in \mathscr{X}_d,\ \phi(x) \text{ defined}\}$$

where $\mathscr{X}_d$ denotes the set of concepts extracted from document d.

The concept-to-cluster mapping is constructed sequentially rather than globally over the complete observation period. For each year t, concepts observed in that year are assigned to stable clusters using only cluster information available up to year t. Once established, these assignments are frozen and are not retrospectively modified using observations from later years. This preserves stable cluster identities across yearly graph construction while preventing future information from influencing historical mappings.

Citation edges For each paper p with reference list $\mathscr{R}(p)$, we create directed edges at cluster level. For each cited paper $r \in \mathscr{R}(p)$, and for each pair of clusters $(C_s, C_t) \in \Phi(p) \times \Phi(r)$ with $C_s \neq C_t$, we add a weighted directed edge $C_s \to C_t$.

To conserve total citation mass when a paper spans multiple clusters, edge contributions are normalized by the number of source clusters:

$$w(C_s, C_t) \mathrel{+}= \frac{1}{|\Phi(p)|}$$

For each year, raw citation weights are accumulated from the available citation network. Yearly graphs are subsequently merged into a cumulative graph by summing raw edge weights over time.:

$$w^{\text{raw}}(C_s, C_t) = \sum_{p} \sum_{r \in \mathscr{R}(p)} \frac{\mathbf{1}[C_s \in \Phi(p),\, C_t \in \Phi(r)]}{|\Phi(p)|}$$

However, not all referenced works can be resolved to cluster assignments. A paper p may cite a paper r that falls outside the dataset $\mathscr{D}$ where $\mathscr{D}$ denotes all documents obtained after the data collection step either because it predates the collection window, belongs to a different field, or was not indexed by OpenAlex at the time of retrieval. In such cases, $\Phi(r)$ is undefined and the corresponding citation edge cannot be constructed.

We therefore restrict the reference list of each paper to documents present in $\mathscr{D}$:

$$\mathscr{R}^*(p) = \{r \in \mathscr{R}(p) \mid r \in \mathscr{D}\}$$

and replace $\mathscr{R}(p)$ with $\mathscr{R}^*(p)$ throughout the graph construction. The raw edge weight formula becomes:

$$w^{\text{raw}}(C_s, C_t) = \sum_{p \in \mathscr{D}} \sum_{r \in \mathscr{R}^*(p)} \frac{\mathbf{1}[C_s \in \Phi(p),\, C_t \in \Phi(r)]}{|\Phi(p)|}$$

This restriction introduces a systematic bias toward intra-corpus citations: edges representing knowledge flows from outside the dataset window are not captured. However, since $\mathscr{D}$ is constructed to be comprehensive within a defined field and time range, we expect $\mathscr{R}^*(p)$ to approximate $\mathscr{R}(p)$ well for the majority of scientifically influential references. Fig 3.4 shows the number of references that fall inside and outside of the dataset. We can see that in 2000, most of the references are outside of the dataset which is normal because most of them cites works from previous year. In 2025, we have a percentage of 45.03% references in dataset.

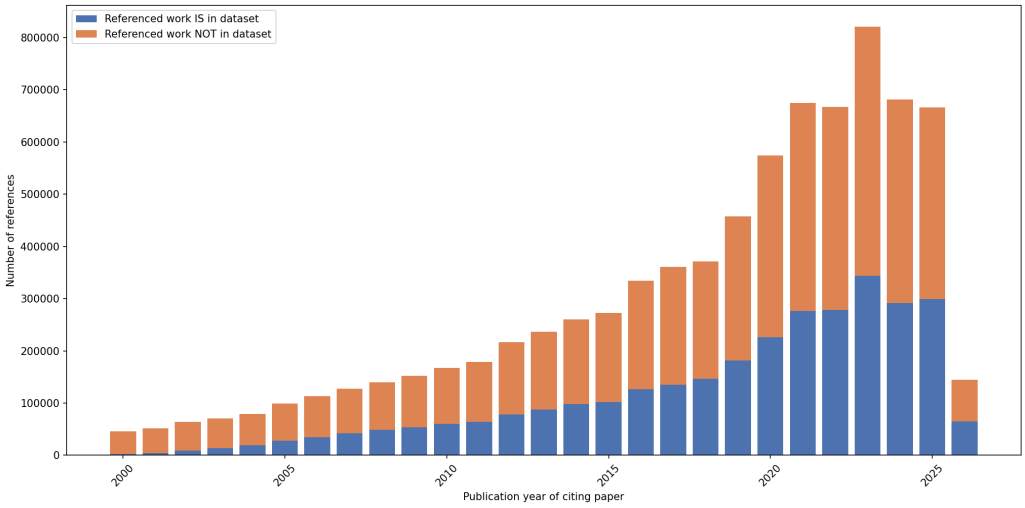


Figure 3.4: References that stay inside vs fall outside the dataset, by year

Patent node boost For each cluster C_k, we aggregate the patent frequency of its member concepts to produce a node-level patent boost signal. Let $\mathrm{pf}(x)$ denote the patent frequency of concept x and $\mathrm{pp}(x)$ its paper frequency. We define:

$$\rho_k = \frac{\sum_{x \in C_k} \mathrm{pf}(x)}{\sum_{x \in C_k} \mathrm{pp}(x) + 1} \quad \text{(patent lead signal)}$$

$$b_k^{\mathrm{node}} = \log\left(1 + \sum_{x \in C_k} \mathrm{pf}(x)\right) \cdot \log(1 + \rho_k)$$

This formulation rewards clusters that are both patent-dense in absolute terms and ahead of papers in relative terms, making b_k^{node} a proxy for technologi-

cal activity that appears earlier or more strongly in patents than in scientific publications.

Patent edge boost Beyond node-level signal, patents that mention concepts from multiple clusters simultaneously provide evidence of cross-cluster technological integration. For each patent q, we count the number of patents co-mentioning concepts from cluster pairs:

$$b^{\text{edge}}(C_s, C_t) = \left|\{q \mid C_s \in \Phi(q) \text{ and } C_t \in \Phi(q)\}\right|$$

This is normalized to $[0,1]$ by dividing by the maximum co-occurrence count across all pairs.

Boosted edge weight. The final edge weight combines the accumulated citation mass with node-level and edge-level patent signals:

$$w(C_s, C_t) = w^{\text{raw}}(C_s, C_t)\big(1 + \alpha_{\text{node}} b_s^{\text{node}}\big)\big(1 + \alpha_{\text{node}} b_t^{\text{node}}\big)\big(1 + \alpha_{\text{edge}} \hat{b}^{\text{edge}}(C_s, C_t)\big).$$

Here, b_s^{node} and b_t^{node} denote the patent-derived node boosts associated with the source and target clusters, respectively, and $\hat{b}^{\text{edge}}(C_s, C_t)$ denotes the normalized patent co-occurrence boost for the corresponding cluster pair. The edge-level boost is normalized within each year by dividing its patent co-occurrence count by the largest observed count in that year.

The coefficients α_{node} and α_{edge} independently control the contributions of node-level and edge-level patent information. Separate coefficients are used because the two signals are derived from different forms of patent evidence and exhibit different numerical distributions. Their values are selected empirically through the sensitivity analysis described below.

Before applying the patent-based weighting, weak yearly citation relationships are removed using a minimum raw-weight threshold. The remaining relationships are further sparsified by retaining only the strongest outgoing edges for each source cluster. This pruning step reduces memory usage and limits the influence of weak, incidental cluster associations while preserving the dominant knowledge-flow relationships.

Concept clustering and stabilization are performed sequentially in chronological order. For each year t, the clusters obtained from that year's concept vocabulary are compared only with stable centroids accumulated from years prior to t. Clusters satisfying the cosine-similarity threshold are assigned an

existing stable identifier, while unmatched clusters receive a new identifier. After assignment, the corresponding stable centroids are updated using the current year's information. The resulting assignments for year t are then saved and are never retrospectively modified when later years are processed.

Consequently, although the stabilization pipeline can be executed chronologically over the complete observation period, the mapping saved for 2022 is identical to the mapping that would be obtained by stopping the sequential procedure at the end of 2022: observations from 2023 onward have not yet been processed when the 2022 assignments are produced.

For the 2022→2025 retrospective validation, the cumulative graph used for forecasting is therefore constructed only from data spanning 2000–2022. All graph, bibliometric, patent-derived, and temporal features used at the forecast origin are obtained from this representation. A second cumulative graph is constructed from 2000–2025 to obtain the realized 2025 network state used for evaluation. Data from 2023–2025 therefore contribute only to the held-out outcome and do not influence the representation used to generate the 2022 forecast.

Algorithm 3 summarizes the yearly graph-construction procedure.

Algorithm 3 Yearly Citation Graph Construction with Patent Boosting

Require: Reference map $\mathscr{R} : p \mapsto \{r_1, r_2, \ldots\}$, weighted document-to-cluster map $\Phi : d \mapsto \{(c, q_{d,c})\}$, patent node boost B^{node}, patent edge boost B^{edge}, coefficients α_{node} and α_{edge}, minimum raw weight τ, maximum outgoing edges K

Ensure: Directed yearly cluster-level citation graph G^t

$raw \leftarrow \{\}$ ▷ Initialise edge-weight accumulator
for all paper p in $\mathscr{R}$ **do**
  $S_p \leftarrow \Phi(p)$
  if $S_p = \emptyset$ **then continue**
  end if
  for all reference r in $\mathscr{R}(p)$ **do**
    $T_r \leftarrow \Phi(r)$
    if $T_r = \emptyset$ **then continue**
    end if
    for all $(c_s, q_{p,c_s}) \in S_p$ **do**
      for all $(c_t, q_{r,c_t}) \in T_r$ **do**
        if $c_s \neq c_t$ **then**
          $raw[c_s \to c_t] \mathrel{+}= q_{p,c_s} q_{r,c_t}$
        end if
      end for
    end for
  end for
end for
Remove all edges (c_s, c_t) satisfying $raw[c_s \to c_t] < \tau$
for all source cluster c_s **do**
  Retain at most the K outgoing edges with the largest raw weights
end for
$b_{\max} \leftarrow \max_{(c_s,c_t)} B^{\text{edge}}(c_s, c_t)$
for all retained edge (c_s, c_t) **do**
  $\hat{b}^{\text{edge}}(c_s, c_t) \leftarrow B^{\text{edge}}(c_s, c_t) / \max(b_{\max}, 1)$
  $w(c_s, c_t) \leftarrow raw[c_s \to c_t]\big(1 + \alpha_{\text{node}} B^{\text{node}}(c_s)\big)\big(1 + \alpha_{\text{node}} B^{\text{node}}(c_t)\big)\big(1 + \alpha_{\text{edge}} \hat{b}^{\text{edge}}(c_s, c_t)\big)$
end for
Add every source and target cluster as a node in G^t
Add every retained directed edge $(c_s \to c_t)$ with raw weight $raw[c_s \to c_t]$ and boosted weight $w(c_s, c_t)$
return G^t

3.3.1 Cumulative Graph and Network Metrics

The knowledge-flow network is constructed cumulatively. For each year t, the yearly citation graph G^t is merged into a running edge accumulator. For an edge that has appeared previously, its yearly raw citation mass is added to its existing cumulative raw weight. Newly observed edges are inserted with their first-observation year, while the most recent patent node and edge attributes are retained. This procedure produces a cumulative graph $G^{\leq t}$ containing all retained citation relationships observed up to year t.

Patent-enhanced cumulative edge weights are recomputed from the accumulated raw citation mass and the most recently available patent attributes:

$$w^{\leq t}(C_s, C_t) = w^{\leq t}_{\text{raw}}(C_s, C_t)\big(1 + \alpha_{\text{node}} b^{\text{node}}_{s,t}\big)\big(1 + \alpha_{\text{node}} b^{\text{node}}_{t,t}\big)\big(1 + \alpha_{\text{edge}} b^{\text{edge}}_{s,t}\big).$$

Thus, citation evidence accumulates over time, whereas patent-derived attributes reflect the most recently observed signal associated with each retained node or edge.

Network metrics are computed directly from the cumulative edge accumulator using sparse matrix operations, avoiding the construction of a multi-million-edge NetworkX graph. For each cluster, the framework computes structural indicators including in-degree, out-degree, PageRank [30], and weighted incoming and outgoing strengths. The primary forecasting target, in-strength, is the sum of the patent-enhanced weights of all incoming edges:

$$k^{\text{in}}_{C_t} = \sum_{C_s \to C_t} w^{\leq t}(C_s, C_t).$$

PageRank is computed using a sparse transition matrix and damping factor $d = 0.85$. Conceptually, the PageRank score π_k of cluster C_k follows the weighted random-walk formulation

$$\pi_k = \frac{1-d}{N} + d \sum_{C_j \to C_k} \frac{w^{\leq t}(C_j, C_k)}{\sum_{C_l} w^{\leq t}(C_j, C_l)},$$

where N denotes the number of clusters in the cumulative graph.

Each cluster is additionally annotated with its mean Field-Weighted Citation Impact and mean citation count, aggregated over the papers whose concepts are assigned to that cluster. These publication-quality indicators complement the network-based measures by distinguishing structurally central fields from

fields supported by highly cited or field-normalized influential publications.

3.4 Forecasting

3.4.1 Forecasting Setup

The forecasting validation pipeline was trained on temporal citation-network data spanning 2000–2022, encompassing 10,021 research clusters across 23 annual snapshots, 22,222,676 inter-cluster edges, 109,875 patent-boost observations, and 251,642 patent-edge records. The model predicts the three-year-ahead logarithmic growth in in-strength. The predicted growth is then combined with the current in-strength to reconstruct the forecasted in-strength at the end of the prediction horizon. In-strength is defined as the sum of boosted incoming edge weights and directly reflects the patent-aware weighting scheme introduced in Section 3.3. A LightGBM gradient-boosted regressor was trained using 105 features derived from node-level metrics (PageRank, in-/out-degree, field-weighted citation impact (FWCI), mean citations per paper, and paper count), edge-level aggregates (incoming and outgoing citation strength, new-edge counts, and boost signals), patent co-occurrence signals, and their lagged and rolling transformations over one- to three-year windows.

To respect temporal ordering and avoid look-ahead bias, a *walk-forward* cross-validation scheme was employed. At each fold i, the model is trained on all years [2000, t_i] and evaluated on its ability to predict in-strength at year $t_i + H$, using only features available at t_i. The forecast origin year t_i ranges from 2005 to 2019, corresponding to prediction targets from 2008 to 2022. For example, fold 0 trains on data up to 2004, constructs features at 2005, and predicts in-strength at 2008.

The transformed target variable has a mean of 0.828 and a standard deviation of 0.657. The correlation between current in-strength and the three-year-ahead log-growth target is -0.319, indicating a moderate negative association. Clusters with higher current in-strength tend to exhibit lower subsequent relative growth, whereas smaller clusters are more likely to experience stronger growth over the forecast horizon. This suggests that predicting future influence cannot rely solely on current network strength and instead requires integrating additional network, bibliometric, and patent-related features to capture the complex dynamics of technological evolution.

The walk-forward cross-validation results are reported in Table 3.7. Across the

15 walk-forward folds, the LightGBM model achieved an average Mean Absolute Error (MAE) of 0.2248 ± 0.0416 and an average Root Mean Squared Error (RMSE) of 0.3640 ± 0.0481 on the transformed prediction target. Prediction accuracy remains relatively stable throughout the evaluation period, with MAE ranging from 0.1806 to 0.3269 and RMSE ranging from 0.3042 to 0.4555. Although the earliest and latest forecast origins exhibit slightly larger errors, no systematic deterioration in performance is observed over time. As expected, RMSE consistently exceeds MAE because it penalizes larger prediction errors more heavily, indicating that a limited number of clusters remain more difficult to forecast than the majority. Overall, the consistently low error values across all walk-forward folds demonstrate that the proposed forecasting model generalizes well to unseen future periods while relying exclusively on information available at each forecast origin.

Table 3.7: Walk-forward cross-validation performance of the LightGBM forecasting model. Each fold evaluates a three-year-ahead prediction using only information available at the corresponding forecast origin year. Performance is reported on the transformed prediction target.

Fold	Forecast origin year	MAE	RMSE
0	2005	0.3269	0.4555
1	2006	0.2955	0.4292
2	2007	0.2600	0.3938
3	2008	0.2440	0.3731
4	2009	0.2098	0.3257
5	2010	0.2066	0.3281
6	2011	0.1943	0.3115
7	2012	0.2050	0.3354
8	2013	0.2025	0.3319
9	2014	0.1942	0.3304
10	2015	0.1806	0.3042
11	2016	0.2056	0.3694
12	2017	0.1999	0.3634
13	2018	0.2018	0.3625
14	2019	0.2457	0.4455
Mean ± Std.	–	0.2248 ± 0.0416	0.3640 ± 0.0481

3.4.2 Scoring and Ranking

The LightGBM model predicts the logarithmic growth in in-strength over a three-year forecasting horizon. The predicted logarithmic growth is subsequently transformed into a forecasted future in-strength, from which the predicted absolute and relative growth are computed. Although these forecasts estimate future citation influence, ranking clusters solely by forecasted future in-strength would naturally favor already large and well-established research areas. To better identify emerging technologies, the forecasting pipeline transforms the model predictions into two complementary ranked lists, denoted Tier 1 and Tier 2, each designed to capture a different emergence pattern.

Both tiers first apply a common set of viability constraints to remove immature or weakly established clusters. Specifically, candidate clusters must possess at least five years of historical observations, contain a minimum of twenty-five papers, exhibit positive absolute growth during the previous three years, and have a non-zero Field-Weighted Citation Impact (FWCI).

Tier-1 Scoring: Rising Academic Fields

Tier 1 aims to identify research areas that are already demonstrating sustained scientific momentum and are expected to experience substantial future growth. Rather than simply selecting the largest research fields, candidate clusters are restricted to those whose current in-strength lies between the 25th and 75th percentiles of the network. This focuses the ranking on medium-sized research areas that have the potential to become future leaders while avoiding both nascent topics with insufficient evidence and already dominant fields.

The Tier 1 score is computed as

$$\begin{aligned} S_1 = \log(1+\widehat{\Delta}) \cdot \log(1+\widehat{r}) \cdot \log(1+\Delta_3) \cdot \log(1+r_3) \\ \cdot \log(1+\mathrm{FWCI}) \cdot \log(1+\bar{c}), \end{aligned} \tag{3.1}$$

where $\widehat{\Delta}$ denotes the predicted absolute in-strength growth over the forecasting horizon, $\widehat{r}$ denotes the predicted relative growth, both derived from the forecasted future in-strength, Δ_3 and r_3 are the observed absolute and relative growth during the preceding three years, respectively, FWCI is the Field-Weighted Citation Impact, and $\bar{c}$ is the mean number of citations per paper.

The multiplicative formulation requires a cluster to simultaneously exhibit strong predicted future growth, sustained historical momentum, and high re-

search quality. Logarithmic transformations are applied to each component to reduce the influence of extreme values while preserving the relative ordering of candidate clusters.

Tier-2 Scoring: Patent-Driven Emerging Technologies

Tier 2 is designed to identify research areas for which predicted scientific growth is accompanied by increasing patent activity. It complements Tier 1, which focuses primarily on scientific emergence, by prioritizing clusters that also exhibit evidence of technological development. Patent information is therefore used as a ranking signal rather than as a replacement for the scientific-growth forecast.

Both tiers restrict candidates according to their current position in the network in order to distinguish emerging areas from already dominant fields. Tier 1 retains clusters whose current in-strength lies between the 25th and 75th percentiles of the network. The lower bound removes very small clusters for which apparent growth may be driven by a low initial value, while the upper bound excludes fields that are already among the most established areas of the network. Tier 2 does not impose the lower bound because patent activity provides an additional signal with which to identify technologically active fields that may still have a relatively small scientific-network presence. It nevertheless applies the same upper 75th-percentile maturity threshold.

In addition to this maturity criterion, Tier 2 requires positive predicted future growth, positive recent patent activity, and positive patent-activity velocity. The eligible set is therefore defined by

$$\widehat{\Delta} > 0, \qquad P > 0, \qquad \dot{P} > 0, \qquad I_t \leq Q_{0.75}(I_t),$$

where $\widehat{\Delta}$ denotes predicted future growth, P denotes recent patent activity, $\dot{P}$ its recent velocity, I_t is current in-strength, and $Q_{0.75}(I_t)$ denotes its 75th percentile at the forecast origin. These conditions enforce consistency between the scientific and technological signals while restricting the ranking to clusters that have not yet reached the upper quartile of current network strength.

For the remaining candidates, the Tier 2 score is defined as

$$\begin{aligned} S_2 = {} & \log(1+\widehat{\Delta}) \cdot \log(1+\widehat{r}) \cdot \log(1+\mathrm{FWCI}) \cdot \log(1+\bar{c}) \\ & \times \left(1+\lambda_P \log(1+P)\right)\left(1+\lambda_{\dot{P}} \log(1+\dot{P})\right)\left(1+\lambda_{\dot{E}} \log(1+\dot{E})\right), \end{aligned} \tag{3.2}$$

where $\hat{r}$ denotes predicted relative growth, FWCI is the mean field-weighted citation impact of the cluster, $\bar{c}$ is its mean citation impact, and $\dot{E}$ denotes the velocity of patent-supported inter-cluster connections.

The first four terms constitute the scientific-emergence component of the score. The three patent quantities are introduced as multiplicative modulators rather than as independent additive components. Each patent term takes the general form

$$1 + \lambda \log(1 + x).$$

This formulation has two motivations. First, the additive constant ensures that patent information modifies rather than replaces the underlying scientific score. Second, the logarithmic transformation reduces the influence of exceptionally large patent signals, whose empirical distributions are strongly skewed. Consequently, clusters must first exhibit scientific potential, after which patent evidence can increase their priority.

The baseline coefficients are

$$\lambda_P = 0.25, \qquad \lambda_{\dot{P}} = 0.25, \qquad \lambda_{\dot{E}} = 0.10.$$

The equal coefficients assigned to P and $\dot{P}$ reflect their complementary interpretation as the current magnitude and recent acceleration of patent activity. A smaller coefficient is assigned to $\dot{E}$ because cross-cluster patent connectivity represents a more indirect indication of technological activity. Importantly, these coefficients are not parameters estimated by LightGBM and should not be interpreted as statistically optimal values. They are moderate heuristic scaling factors used in the post-forecast ranking procedure.

Because manually specified weights may introduce sensitivity into the resulting ranking, we evaluated alternative coefficient combinations while keeping the forecasting model, candidate filters, and validation period fixed. Table 3.8 summarizes the results for the 2022–2025 retrospective validation.

The sensitivity analysis indicates that the emergence framework is relatively robust to the choice of patent-boost coefficients in terms of candidate selection and realized growth percentiles. Across the tested configurations, the mean global growth percentile of the top-10 candidates varies only from 95.1% to 95.8%, while NDCG@10 (Normalized Discounted Cumulative Gain) remains between 0.926 and 0.957. No single coefficient configuration dominates all evaluation metrics: for example, the largest Spearman correlation is obtained with

Table 3.8: Sensitivity of Tier 2 validation performance to the patent-weight coefficients for the 2022–2025 forecast.

λ_P	$\lambda_{\dot{P}}$	$\lambda_{\dot{E}}$	Top-10 percentile	Spearman	NDCG@10
0.05	0.05	0.05	95.1%	0.669	0.926
0.10	0.10	0.10	95.6%	0.690	0.931
0.15	0.15	0.05	95.6%	0.662	0.929
0.20	0.20	0.05	95.6%	0.681	0.931
0.25	**0.25**	**0.10**	**95.8%**	0.680	0.946
0.40	0.40	0.10	95.8%	0.657	**0.957**
0.40	0.40	0.20	95.8%	0.638	0.954

$(0.10, 0.10, 0.10)$, whereas the largest NDCG@10 is obtained with $(0.40, 0.40, 0.10)$. The baseline $(0.25, 0.25, 0.10)$ is therefore retained as an intermediate specification rather than as a retrospectively optimized parameter combination.

The same coefficient analysis was repeated for an earlier 2019–2022 forecasting window to examine whether the conclusions were specific to the final validation period. Candidate-selection performance remained comparable under alternative coefficient settings, with the mean top-10 global percentile remaining approximately between 93.6% and 94.7%. This provides additional evidence that the Tier 2 results are not driven by a narrow set of coefficients chosen specifically for the 2022–2025 evaluation.

The upper in-strength threshold is also a heuristic component of the emergence definition. The default 75th percentile is intended to exclude the most established quarter of the network rather than to represent a theoretically unique boundary between emerging and mature fields. We therefore repeated the 2022–2025 validation while varying the upper threshold from the 60th to the 90th percentile and keeping the patent coefficients fixed at $(0.25, 0.25, 0.10)$. For Tier 1, the lower bound remained fixed at the 25th percentile while its upper bound was varied simultaneously. For Tier 2, only the upper maturity bound was varied.

The sensitivity analysis reveals a trade-off rather than a sharply optimal threshold. For Tier 1, increasing the upper bound allows larger and more established clusters to enter the candidate set and raises the mean global growth percentile of the top-10 predictions from 96.1% at the 60th percentile to 98.4% at the 90th percentile. At the same time, NDCG@10 decreases from 0.983 to 0.825. Thus, a more permissive threshold can retrieve clusters with very large subsequent absolute growth, but does not necessarily improve the quality of their ordering.

Table 3.9: Sensitivity of the 2022–2025 validation to the upper current in-strength threshold. Patent coefficients are fixed at $(0.25, 0.25, 0.10)$.

Upper threshold	Tier 1 Top-10	Tier 1 NDCG@10	Tier 2 Top-10	Tier 2 NDCG@10
60%	96.1%	0.983	93.7%	0.949
65%	96.1%	0.983	93.7%	0.949
70%	96.2%	0.891	94.8%	0.938
75%	97.1%	0.897	95.8%	0.946
80%	98.1%	0.843	95.8%	0.946
85%	98.1%	0.843	95.9%	0.936
90%	98.4%	0.825	95.9%	0.936

Tier 2 exhibits similar robustness in candidate retrieval. Its mean top-10 global growth percentile ranges from 93.7% at the 60th percentile to approximately 95.9% at the 85th–90th percentiles, and all top-10 candidates remain above the 75th percentile of realized global growth throughout the experiment. NDCG@10 remains between 0.936 and 0.949. Thus, the principal conclusion that Tier 2 retrieves clusters with unusually strong subsequent growth is preserved over a broad range of maturity thresholds.

We retain the 75th percentile as the baseline upper bound because it provides an interpretable compromise between two competing objectives. A substantially lower threshold imposes a restrictive definition of emergence and removes fields that have already developed a moderate scientific presence, whereas a higher threshold increasingly admits established clusters whose large absolute growth may reflect their existing scale. The 75th-percentile cutoff excludes the upper quartile of the network while retaining a sufficiently broad candidate population. Importantly, it is not selected because it maximizes a particular retrospective validation metric; the sensitivity analysis instead shows that the main conclusions remain stable under alternative choices.

Overall, these experiments indicate that Tier 2 is not dependent on a narrowly tuned combination of patent weights or maturity thresholds. The heuristic parameters influence the precise ordering and composition of the candidate set, as expected, but high-growth candidates are recovered across a broad range of specifications. The Tier 2 score should nevertheless be interpreted as an operational ranking mechanism rather than a fully learned decision rule. Future work could replace these manually specified parameters with coefficients estimated across multiple historical forecasting windows or technological domains.

The complete emergence-detection procedure is summarized in Algorithm 4. Following the LightGBM prediction, candidate clusters are first filtered using

the common viability criteria. The remaining clusters are then separated into two complementary rankings. Tier 1 identifies scientifically emerging fields by combining predicted future growth, historical momentum, and research quality indicators. Tier 2 further emphasizes those growing fields that also display increasing industrial activity through patent-derived signals. The highest-ranked clusters from each tier are finally returned as the predicted emerging research areas.

Algorithm 4 Two-Tier Emergence Detection

Require: Forecast predictions $\hat{Y}$, feature matrix $\mathbf{F}$, patent indicators $\mathscr{P}$, top-N parameter

Ensure: Tier 1 rising academic fields $\mathscr{T}_1$, Tier 2 patent-informed emerging technologies $\mathscr{T}_2$

— Candidate Filtering —

$\mathscr{B} \leftarrow$ clusters with at least five years of historical observations

$\mathscr{B} \leftarrow \mathscr{B} \cap \{c \mid \overline{\text{FWCI}}_c > 0,\ \overline{\text{cit}}_c > 0,\ n_{\text{papers},c} \geq 25\}$ ▷ Quality and viability filter

for all $c \in \mathscr{B}$ **do**

  $\Delta^{\text{abs}}_{3,c} \leftarrow k^{\text{in}}_{c,T} - k^{\text{in}}_{c,T-3}$

  $\Delta^{\text{rel}}_{3,c} \leftarrow \Delta^{\text{abs}}_{3,c} / (k^{\text{in}}_{c,T-3} + \varepsilon)$

  $\widehat{\Delta}_c \leftarrow \hat{k}^{\text{in}}_{c,T+h} - k^{\text{in}}_{c,T}$

  $\widehat{r}_c \leftarrow \widehat{\Delta}_c / (k^{\text{in}}_{c,T} + \varepsilon)$

end for

$\mathscr{B} \leftarrow \mathscr{B} \cap \{c \mid \Delta^{\text{abs}}_{3,c} > 0\}$

$p_{25} \leftarrow Q_{0.25}(k^{\text{in}}_{c,T}), \quad p_{75} \leftarrow Q_{0.75}(k^{\text{in}}_{c,T})$

— Tier 1: Rising Academic Fields —

$\mathscr{T}_1 \leftarrow \{c \in \mathscr{B} \mid p_{25} \leq k^{\text{in}}_{c,T} \leq p_{75}\}$

for all $c \in \mathscr{T}_1$ **do**

  $\text{score}_1(c) \leftarrow \ln(1+\max(\widehat{\Delta}_c, 0)) \cdot \ln(1+\max(\widehat{r}_c, 0)) \cdot \ln(1+\max(\Delta^{\text{abs}}_{3,c}, 0)) \cdot \ln(1+\max(\Delta^{\text{rel}}_{3,c}, 0)) \cdot \ln(1+\overline{\text{FWCI}}_c) \cdot \ln(1+\overline{\text{cit}}_c)$

end for

— Tier 2: Patent-Informed Emerging Technologies —

$\mathscr{T}_2 \leftarrow \{c \in \mathscr{B} \mid P_{c,T} > 0 \wedge \dot{P}_{c,T} > 0 \wedge \widehat{\Delta}_c > 0 \wedge k^{\text{in}}_{c,T} \leq p_{75}\}$

for all $c \in \mathscr{T}_2$ **do**

  $\text{score}_2(c) \leftarrow \ln(1+\max(\widehat{\Delta}_c, 0)) \cdot \ln(1+\max(\widehat{r}_c, 0)) \cdot \ln(1+\overline{\text{FWCI}}_c) \cdot \ln(1+\overline{\text{cit}}_c) \cdot [1 + 0.25\ln(1 + \max(P_{c,T}, 0))] \cdot [1 + 0.25\ln(1 + \max(\dot{P}_{c,T}, 0))] \cdot [1 + 0.10\ln(1 + \max(\dot{E}^{\text{pat}}_{c,T}, 0))]$

end for

$\mathscr{T}_1 \leftarrow \mathscr{T}_1$ sorted in descending order by score_1

$\mathscr{T}_2 \leftarrow \mathscr{T}_2$ sorted in descending order by score_2

return $\mathscr{T}_1[1:N], \quad \mathscr{T}_2[1:N]$

Chapter 4

Validation and Results

4.1 Validation

Validation is performed retrospectively using a three-year forecasting horizon. The model is trained using information available up to 2022, and predictions generated at the 2022 cutoff are compared with the growth subsequently observed in 2025. Only clusters that can be consistently identified at both the forecast origin and the evaluation year are retained for evaluation. This results in 10,599 matched clusters from the 10,784 clusters available at the 2022 cutoff and the 14,367 clusters observed in 2025.

Validation is structured at two levels. First, forecast validity is assessed by comparing the predicted ranking of candidate clusters at the 2022 cutoff with their realized three-year growth in in-strength in 2025. Performance is evaluated using Spearman rank correlation, NDCG@10, the mean global percentile of the top-10 candidates, and growth lift. These metrics evaluate complementary properties of the forecast, including ranking consistency, top-ranked candidate quality, and the magnitude of subsequently observed growth.

Second, the contribution of patent information is evaluated through an ablation experiment. The complete forecasting pipeline is rerun after removing patent-derived node and edge boosts as well as patent-derived model features. This reduces the feature space from 105 features in the full model to 61 features in the publication-only configuration. Both configurations are evaluated using the same 2022–2025 forecast horizon and, where applicable, the same set of eligible clusters, ensuring that differences in performance can be attributed to the inclusion of patent information rather than to differences in the evaluation population.

4.1.1 In-Strength versus In-Degree as the Forecasting Target

The primary forecasting target used in this study is *in-strength*, defined as the sum of the boosted incoming edge weights associated with each cluster in the knowledge-flow network. Unlike *in-degree*, which counts only the number of distinct incoming links, in-strength incorporates the weights of these relationships and therefore reflects both scientific citation activity and the patent-derived signals introduced during graph construction. The main forecasting experiments use a three-year horizon ($H = 3$), while additional one- and five-year horizons are considered here to evaluate the choice of forecasting target.

To determine whether weighted in-strength provides additional predictive information beyond simple network connectivity, in-degree is used as an unweighted structural baseline. Both targets were evaluated under identical conditions across 11 valid combinations of forecast cutoff years and horizons, with $t_i \in \{2016, 2018, 2020, 2022\}$ and $H \in \{1, 3, 5\}$, excluding configurations extending beyond the available observation period. Network construction, feature engineering, forecasting, and ranking procedures were kept unchanged, with only the forecasting target being modified. Performance was evaluated against held-out future observations using ranking and realized-growth metrics.

Table 4.1 summarizes the comparison between the two forecasting targets.

Table 4.1: Comparison of in-degree and in-strength forecasting targets across the 11 temporal validation configurations.

Tier	**Metric**	**In-degree**	**In-strength**
Tier 1	Mean top-10 growth percentile	96.69%	**97.29%**
	Mean Spearman correlation	0.749	**0.791**
	Configurations with higher percentile	2/11	**9/11**
Tier 2	Mean top-10 growth percentile	93.31%	**93.33%**
	Mean Spearman correlation	0.638	**0.727**
	Mean NDCG@10	0.853	**0.876**
	Top-10 above 75th percentile	**93.6%**	92.7%
	Configurations with higher percentile	3/11	**8/11**

For Tier 1, in-strength achieves a mean top-10 global growth percentile of 97.29%, compared with 96.69% for in-degree, and increases the mean Spearman rank correlation from 0.749 to 0.791. In-strength also achieves the higher mean growth percentile in 9 of the 11 temporal configurations. Although the difference in absolute percentile is relatively small, both targets already identify clusters near the upper end of the realized growth distribution. The improvement in

rank correlation therefore indicates that in-strength provides a more consistent ordering of scientifically emerging candidates.

For Tier 2, the two targets identify candidates with nearly identical realized growth percentiles, with mean values of 93.31% for in-degree and 93.33% for in-strength. The proportion of top-10 candidates above the 75th percentile is also similar, reaching 93.6% and 92.7%, respectively. However, in-strength provides stronger ranking performance within the Tier 2 candidate population: the mean Spearman correlation increases from 0.638 to 0.727, while mean NDCG@10 increases from 0.853 to 0.876. In-strength also achieves the higher mean growth percentile in 8 of the 11 configurations. These results suggest that its main advantage for Tier 2 lies in the prioritization of high-growth candidates rather than in their initial detection.

Across both tiers, in-strength achieves a higher mean global growth percentile in 17 of the 22 target–tier comparisons (77.3%). Overall, the results indicate that weighted in-strength provides a modest but consistent improvement over unweighted in-degree, particularly in ranking candidates according to their subsequently realized growth. Its stronger ranking performance, together with its ability to incorporate the patent-aware weighting of the knowledge-flow network, motivates its use as the primary forecasting target throughout the remainder of this study. In-degree is retained as an unweighted structural baseline.

4.1.2 Feature Importance

Table 4.2 reports the twenty most important predictors according to LightGBM's split-based feature-importance measure. The highest-ranked features are the current mean citation count, PageRank, and mean Field-Weighted Citation Impact (FWCI), indicating that both research quality and structural prominence in the citation network provide strong signals of subsequent in-strength growth. The patent-related variable *boost sent* is the fourth most important feature, while *boost received* also appears among the ten highest-ranked predictors. These results suggest that patent-aware information contributes meaningful predictive value, although it operates alongside rather than above the principal bibliometric and network indicators.

Measures of citation flow are also prominent in the ranking. Current outgoing and incoming strength, outgoing degree, and raw outgoing strength all appear among the most influential predictors, showing that both weighted network

activity and unweighted connectivity help characterize future cluster development. Temporal transformations further contribute to the forecast: three-year rolling citation and FWCI averages, PageRank velocity, strength velocity, lagged citation values, and citation-impact velocity all appear in the top twenty. Overall, the feature ranking indicates that future in-strength growth is best predicted through a combination of current scientific impact, network position, citation-flow intensity, temporal momentum, and patent-aware signals.

Table 4.2: Top-20 features by LightGBM split-based importance.

Feature	Importance
mean_citations	1389
pagerank	1363
mean_fwci	1343
boost_sent	1281
out_strength	987
in_strength	941
out_degree	885
mean_citations_roll3	845
boost_received	822
n_papers	803
out_raw_strength	782
pagerank_vel	638
mean_fwci_roll3	619
in_raw_strength	536
out_strength_vel	524
in_strength_vel	504
mean_citations_lag1	502
mean_citations_vel	472
mean_fwci_vel	471
pagerank_roll3	447

4.1.3 Selection of Patent Boost Coefficients.

Before tuning the patent-boost coefficients, we examined the coverage of the node and edge-level patent signals in the constructed citation graphs. The node-level boost, b_{node}, became progressively more prevalent over the observation period. The proportion of nodes with a nonzero patent boost increased from 12.08% in 2000, corresponding to 90 of 745 nodes, to 51.18% in 2022, corre-

sponding to 4,877 of 9,530 nodes. Over the same period, the number of patents contributing to the patent co-occurrence signal increased from 464 to 8,402.

In contrast, the edge-level boost, b_{edge}, remained extremely sparse throughout the observation period because it requires a retained citation edge to connect two clusters that are also co-mentioned in at least one patent. In 2000, only 6 of 2,242 retained citation edges received a nonzero edge boost, corresponding to 0.2676%. By 2022, 488 of 451,729 retained edges were boosted, corresponding to 0.1080%. Across most years, edge-level coverage remained below 0.1%, despite the substantial growth in both patent activity and network size.

This difference in coverage is also reflected in the forecasting model. Among the 105 engineered features used by the LightGBM predictor, the node-derived variables *boost_sent* and *boost_received* ranked fourth and ninth, respectively, indicating that node-level patent information contributes meaningful predictive value. In contrast, no feature directly derived from the edge-level patent boost appeared among the twenty highest-ranked predictors. These results suggest that patent information contributes primarily through broad node-level signals, whereas the very limited coverage of patent-enhanced citation edges constrains the additional predictive value of the edge-level boost.

To determine suitable values for the patent-boost coefficients $(\alpha_{\text{node}}, \alpha_{\text{edge}})$, we performed a grid search over

$$\alpha_{\text{node}} \in \{0.1, 0.3, 0.5, 0.7, 1.0\}, \qquad \alpha_{\text{edge}} \in \{0.1, 0.2, 0.3, 0.5\}.$$

The prediction target was fixed to in-strength with a forecasting horizon of $H = 3$, while all other components of the forecasting and ranking pipeline were kept unchanged. During graph construction, each source cluster was restricted to its 100 strongest outgoing citation edges, so that all coefficient combinations were evaluated under the same top-100 edge-cap configuration. Each pair of coefficients was assessed using held-out future growth of the clusters returned by the Tier 1 and Tier 2 ranking procedures. The evaluation focused on the mean and median global growth percentiles of the top-ranked clusters, the Spearman correlation between predicted ranking scores and realized future growth, NDCG, and lift relative to the mean growth of the complete cluster population.

The coefficient sweep indicates that the node-level coefficient has a stronger effect on the resulting rankings than the edge-level coefficient. Low values of α_{node} generally produced the strongest overall performance, whereas larger values tended to reduce the correlation between the ranking scores and realized

growth. In particular, increasing α_{node} beyond 0.5 did not improve the global percentile of the selected clusters and often reduced ranking quality, suggesting that excessive node-level patent weighting can distort the underlying citation-network signal.

For a fixed value of α_{node}, varying α_{edge} produced only modest changes in the global growth percentiles of the selected candidates. This stability is consistent with the limited coverage of the edge-level patent signal. Nevertheless, some variation was observed in Spearman correlation and NDCG, particularly for Tier 2, indicating that the edge coefficient can affect the internal ordering of patent-informed candidates even when it has little influence on their overall growth percentile.

Table 4.3 shows the result obtained with different values and the highest composite validation score was obtained with

$$\alpha_{\text{node}} = 0.1, \qquad \alpha_{\text{edge}} = 0.1.$$

Under this configuration, the top-10 Tier 1 candidates achieved a mean global growth percentile of 96.9% and a median of 97.0%, with all selected clusters located above the 75th percentile of global future growth. The Tier 1 ranking score exhibited a Spearman correlation of 0.697 with realized growth and achieved NDCG values of 0.903 at 10 and 0.914 at 25.

Tier 2 displayed similarly strong validation performance. Its top-10 candidates achieved a mean global growth percentile of 95.6% and a median of 99.0%, with all selected clusters exceeding the 75th percentile. The corresponding Spearman correlation with realized growth was 0.645, while NDCG reached 0.962 at 10 and 0.954 at 25. The mean realized growth of the top-10 Tier 1 and Tier 2 candidates was approximately 7.98 and 7.94 times the population mean, respectively, indicating substantial enrichment over an unranked selection of research clusters.

The robustness of the selected coefficients was also evaluated using the top-25 candidates. For $\alpha_{\text{node}} = 0.1$ and $\alpha_{\text{edge}} = 0.1$, the Tier 1 top-25 candidates achieved a mean global growth percentile of 95.1%, while the Tier 2 candidates achieved 92.0%. All Tier 1 candidates and 96% of Tier 2 candidates remained above the 75th percentile. The corresponding lifts over the population mean were 5.29 for Tier 1 and 4.95 for Tier 2. These results indicate that the strong top-10 performance is not driven by only a small number of exceptional clusters.

Overall, the sweep shows that relatively conservative patent weighting pro-

vides the best balance between citation-network structure and patent-derived information. We therefore adopt $\alpha_{\text{node}} = 0.1$ and $\alpha_{\text{edge}} = 0.1$ for the final model. This configuration achieved the highest composite score, strong global percentile performance in both tiers, high NDCG, and substantial growth lift, while avoiding the deterioration observed under stronger node-level boosting.

Table 4.3: Patent-boost coefficient sweep using an in-strength forecasting target ($H = 3$) and a top-100 outgoing-edge cap.

α node	α edge	Tier 1 Mean (%)	Tier 1 Spearman	Tier 1 NDCG@10	Tier 2 Mean (%)	Tier 2 Spearman	Tier 2 NDCG@10
0.1	**0.1**	**96.9**	**0.697**	**0.903**	**95.6**	**0.645**	**0.962**
0.5	0.1	96.6	0.653	0.882	95.7	0.695	0.978
0.5	0.2	96.6	0.660	0.880	94.9	0.674	0.985
0.5	0.5	96.6	0.650	0.858	94.9	0.685	0.988
1.0	0.2	96.5	0.509	0.921	95.1	0.655	0.986
0.1	0.2	96.9	0.661	0.823	95.3	0.643	0.943
0.5	0.3	96.6	0.667	0.858	96.0	0.690	0.884
0.7	0.5	96.6	0.579	0.918	95.1	0.669	0.865
0.3	0.1	96.1	0.561	0.759	95.8	0.674	0.965
0.3	0.3	96.1	0.548	0.709	94.7	0.652	0.915

The coefficient sweep in Table 4.3 was conducted under a top-100 outgoing-edge cap for computational tractability. For the final held-out validation and all subsequent results, the edge cap was relaxed to top-150, which accounts for the small differences between the sweep's reported metrics and those in Section 4.1.4

4.1.4 Held-Out Validation Against 2025 Observations

To evaluate the practical utility of the proposed forecasting framework, we performed a held-out validation by training the forecasting model using data available up to 2022 and comparing the predictions against the *actual* state of the citation network in 2025. The objective is not merely to predict future network metrics accurately, but to determine whether the framework can identify research topics that subsequently experience exceptional scientific growth.

The primary prediction target is the future *in-strength* of each concept cluster, which captures the total incoming citation influence accumulated by a research area. Validation therefore compares the predicted 2025 in-strength against the observed in-strength measured in the held-out 2025 network.

To evaluate forecasting quality from multiple perspectives, four complementary criteria are employed.

- **Global growth percentile** measures how the observed growth of each predicted cluster ranks relative to all clusters in the network.
- **Spearman rank correlation** evaluates whether the predicted ranking agrees with the observed ordering of future growth.
- **Normalized Discounted Cumulative Gain (NDCG)** evaluates the quality of the ranking while placing greater emphasis on the highest-ranked predictions.
- **Growth lift** compares the average growth of the predicted clusters with the average growth across the entire network.

Together these metrics assess both the ability of the framework to identify future high-growth research areas and the quality of their ranking.

Tier-1 Validation: Rising Academic Fields

Tier 1 aims to identify research areas that exhibit sustained scientific momentum and are expected to experience strong future growth based primarily on citation-network structure, bibliometric quality, and recent publication dynamics.

The held-out evaluation demonstrates that the proposed forecasting framework successfully identifies rapidly expanding research fields. The top-10 Tier 1 predictions achieved a **mean global growth percentile of 97.1%** and a median percentile of 97.1%. The lowest-ranked cluster in this set was still located at the 94.2nd percentile, and all ten predictions exceeded the 75th percentile of network-wide growth. Extending the evaluation to the top-25 predictions produced a mean global percentile of 95.2% and a median of 95.0%. All 25 candidates remained above the upper quartile of the global growth distribution.

Beyond selecting high-growth clusters, the Tier 1 score also provides a meaningful ordering of the candidates. Across the 30 returned clusters, the Spearman rank correlation between the predicted ranking score and realized future growth was

$$\rho = 0.718, \qquad p = 7.96 \times 10^{-6},$$

indicating strong agreement between the predicted and observed rankings. The ranking achieved an **NDCG@10 of 0.897**, increasing to 0.910 at 25 and 0.912 across all 30 candidates. These results indicate that the highest-scoring Tier 1 recommendations tend to correspond closely to the fields that subsequently

experience the strongest observed growth.

From a practical perspective, the top-10 Tier 1 candidates achieved a mean observed in-strength growth of 417.8, corresponding to a **7.85-fold lift** relative to the average growth of the complete cluster population. The lift remained 5.27 for the top-25 candidates and 4.81 across all 30 candidates, demonstrating that the method consistently concentrates rapidly growing research fields near the top of the ranking.

Table 4.4: Tier 1 validation using held-out 2025 observations. k_{2022} denotes the observed in-strength in 2022, $\widehat{k}_{2025}$ the forecasted in-strength for 2025, $\widehat{\Delta}$ the predicted growth, and Δ the observed growth between 2022 and 2025.

ID	Research topic	k_{2022}	$\widehat{k}_{2025}$	$\widehat{\Delta}$	Δ	Global percentile
5349	Key distribution QKD	307.8	1019.6	711.8	483.0	99.00
4668	Quantum system	129.1	736.4	607.2	1066.5	99.84
2072	Variational quantum eigensolver	352.2	913.8	561.6	636.5	99.44
2782	Quantum simulation	231.5	877.1	645.6	620.6	99.40
2002	Noisy intermediate-scale quantum	152.4	386.1	233.6	199.5	95.24
5351	Control quantum system	240.5	696.6	456.1	418.2	98.62
4938	Quantum body scar	127.5	287.5	159.9	181.8	94.52
4477	Symmetry-protected topological phases	222.4	552.0	329.6	215.9	95.67
2851	Quantum sensor	167.8	406.0	238.2	182.7	94.57
3318	Open quantum system	258.6	709.1	450.5	173.6	94.21

The predicted growth magnitudes are not expected to reproduce the realized values exactly, because the principal objective is to identify and rank high-growth clusters rather than to provide perfectly calibrated point forecasts. Nevertheless, the selected fields consistently underwent substantial subsequent expansion. In particular, *Quantum system*, *Variational quantum eigensolver*, *Quantum simulation*, and *Key distribution QKD* all ranked within approximately the top 1% of network-wide growth between 2022 and 2025. These results provide evidence that the forecasting framework can identify research directions that subsequently exhibit substantial growth within the evaluated quantum-technology corpus.

Tier-2 Validation: Patent-Driven Emerging Technologies

Tier 2 incorporates patent-derived information to identify research areas that combine forecasted scientific growth with increasing industrial momentum. Rather than replacing Tier 1, this ranking is intended to highlight clusters for which patent activity provides an additional leading indicator of future scientific importance.

The held-out evaluation shows that the patent-informed ranking also performs strongly. The top-10 Tier 2 predictions achieved a mean global growth percentile of 95.8% and a median percentile of 98.9%. The lowest-ranked candidate was located at the 81.8th percentile, and all ten predictions exceeded the 75th percentile of network-wide growth. Across the top-25 candidates, the mean global percentile remained high at 92.1%, with a median of 94.5%. In this larger set, 96% of the candidates remained above the 75th percentile.

The Tier 2 ranking score also showed strong agreement with realized future growth. Across the 30 returned candidates, the Spearman correlation was

$$\rho = 0.680, \qquad p = 3.56 \times 10^{-5}.$$

The ranking achieved a NDCG@10 of 0.946, with corresponding values of 0.942 at 25 and 0.941 across all 30 candidates. These results indicate that the patent-informed score effectively prioritizes the candidates that subsequently exhibit the strongest growth.

The top-10 Tier 2 candidates achieved a mean observed in-strength growth of 405.4, corresponding to a 7.62-fold lift over the population mean. The lift remained 4.85 for the top-25 candidates and 4.31 across all 30 candidates. Thus, the Tier 2 ranking identifies a strongly enriched set of future high-growth research areas while incorporating information about industrial activity.

Importantly, Tier 2 does not merely reproduce the citation-driven Tier 1 ranking. Only four clusters were shared between the two top-10 lists, corresponding to an overlap of 40% and a Jaccard similarity of 0.25. Consequently, Tier 2 contributed additional patent-informed candidates. These included *Quantum error correction*, *Quantum machine learning*, *Quantum computational advantage*, and *Neural-network quantum*. At the top-25 cutoff, 12 clusters were shared, leaving 13 Tier 2-specific candidates.

Overall, the held-out evaluation indicates that patent-derived information provides complementary value rather than simply duplicating citation-based forecasting. Tier 1 captures fields with strong citation-network momentum,

Table 4.5: Tier 2 validation using held-out 2025 observations. k_{2022} denotes the observed in-strength in 2022, b the recent patent boost, $\dot{b}$ the patent-boost velocity, and Δ the observed in-strength growth between 2022 and 2025.

ID	Research topic	k_{2022}	b	$\dot{b}$	Δ	Global percentile
5474	Quantum error correction	71.0	0.073	0.022	599.2	99.35
5547	Quantum key distribution QKD	81.7	0.970	0.970	688.5	99.52
5471	Quantum machine learning	59.4	0.051	0.051	438.3	98.74
2072	Variational quantum eigensolver	352.2	0.009	0.009	636.5	99.44
2002	Noisy intermediate-scale quantum	152.4	0.017	0.017	199.5	95.24
5542	Quantum computational advantage	114.1	0.048	0.048	180.2	94.46
5382	Neural-network quantum	27.8	0.215	0.215	78.4	81.82
2782	Quantum simulation	231.5	0.061	0.061	620.6	99.40
759	Machine learning	179.6	2.345	2.345	129.5	91.14
5349	Key distribution QKD	307.8	0.444	0.444	483.0	99.00

whereas Tier 2 highlights additional research areas whose industrial activity precedes or accompanies subsequent scientific expansion. The limited overlap between the two rankings, together with their high global percentiles, rank correlations, NDCG values, and growth lifts, suggests that the two-tier framework increases the diversity of detected emerging technologies without substantially reducing validation performance.

4.1.5 Patent Ablation Analysis

Patent-derived information enters the proposed framework at three distinct stages. First, patent activity is used to construct forecasting features such as the patent-related node and edge signals. Second, patent information is used to reweight the scientific citation graph through the node- and edge-level boosting mechanisms described in Section 3.3. Third, the primary forecasting target, in-strength, is computed from these patent-weighted edges. Consequently, simply removing patent information from the complete pipeline changes not only the model inputs, but also the graph representation and potentially the definition of the forecasting target.

To distinguish these effects, the ablation analysis progressively introduces the different uses of patent information. Four configurations are considered. The *publication-only* configuration uses the raw scientific citation graph, excludes all patent-derived forecasting features, and predicts raw citation in-strength. The *patent-features* configuration retains the same raw graph and raw in-strength target but introduces patent-derived forecasting features. The *patent-weighted graph* configuration additionally applies the patent node and edge boosts to the graph while still predicting raw citation in-strength. Finally, the *full model* uses patent-derived features, the patent-weighted graph, and patent-enhanced in-strength as the forecasting target. This progressive design makes it possible to distinguish the contribution of patent information as a predictive input from its contribution to the graph representation and target definition.

Table 4.6: Ablation configurations used to distinguish the different roles of patent-derived information.

Configuration	**Patent features**	**Graph**	**Target**
Publication-only	No	Raw citation	Raw in-strength
Patent features	Yes	Raw citation	Raw in-strength
Patent-weighted graph	Yes	Patent-weighted	Raw in-strength
Full model	Yes	Patent-weighted	Patent-enhanced in-strength

All configurations are evaluated retrospectively using the same three-year forecasting horizon from 2022 to 2025. The Tier 1 ranking is used for the comparison because Tier 2 explicitly incorporates patent activity in its ranking formulation and therefore does not provide a publication-only counterpart. Performance is evaluated using the mean global growth percentile of the top-10 candidates, Spearman rank correlation between the ranking score and subsequently observed growth, NDCG@10, and top-10 growth lift.

Table 4.7: Tier 1 ablation results for the different uses of patent-derived information in the forecasting pipeline.

Configuration	**Mean percentile**	**Spearman**	**NDCG@10**	**Growth lift**
Publication-only	93.8%	0.399	0.802	5.23×
Patent features	96.9%	0.716	0.889	7.37×
Patent-weighted graph	**97.1%**	**0.740**	0.882	7.44×
Full model	**97.1%**	0.718	**0.897**	**7.85**×

The largest improvement is observed when patent-derived information is introduced as forecasting features while the graph and target remain based on raw

scientific citations. Relative to the publication-only configuration, adding patent features increases the mean global percentile of the top-10 candidates from 93.8% to 96.9% and the Spearman correlation from 0.399 to 0.716. NDCG@10 increases from 0.802 to 0.889, while the top-10 growth lift increases from 5.23× to 7.37×. Because the graph and target are identical in these two configurations, this comparison provides the clearest evidence that patent-derived variables contain complementary predictive information beyond conventional bibliometric and citation-network features.

The additional contribution of patent-based graph weighting is more modest. When patent features and the raw in-strength target are held fixed, replacing the raw citation graph with the patent-weighted graph increases the mean top-10 percentile from 96.9% to 97.1% and the Spearman correlation from 0.716 to 0.740. Growth lift also increases slightly from 7.37× to 7.44×. However, NDCG@10 decreases slightly from 0.889 to 0.882. Patent-based graph weighting therefore does not uniformly improve all evaluation metrics. Instead, it appears to provide a smaller and metric-dependent refinement compared with the larger improvement obtained from introducing patent-derived forecasting features.

The final configuration additionally changes the forecasting target from raw citation in-strength to patent-enhanced in-strength. Compared with the patent-weighted graph configuration using the raw target, the full model increases NDCG@10 from 0.882 to 0.897 and growth lift from 7.44× to 7.85×, while the mean global percentile remains approximately 97.1%. In contrast, Spearman correlation decreases from 0.740 to 0.718. The patent-enhanced target therefore does not provide a uniform improvement across all ranking metrics, but it produces stronger concentration of high-growth clusters near the top of the ranking, as reflected by NDCG@10 and growth lift.

Overall, the ablation analysis indicates that the contribution of patent information should not be interpreted as a single effect. Patent-derived forecasting features account for the largest observed improvement over the publication-only baseline, suggesting that patent activity provides complementary information for distinguishing future high-growth research areas. Patent-based graph weighting provides a smaller additional contribution, while the patent-enhanced target primarily affects the quality of the highest-ranked candidates rather than consistently improving the overall rank correlation. These results therefore support the use of patent information as a complementary refinement mechanism, while avoiding the stronger claim that patent signals are necessary for detecting emerging research areas.

4.2 Results

Following model selection and retrospective validation, the complete forecasting pipeline was retrained using all fully observed annual snapshots available up to 2025 in order to generate prospective predictions for 2028. More recent publication and citation records were excluded because incomplete indexing and citation accumulation could introduce temporal bias into the final network snapshot.

The final forecast uses a three-year horizon, $H = 3$, with in-strength as the principal forecasting quantity and the selected patent-boost coefficients

$$\alpha_{\text{node}} = 0.1, \qquad \alpha_{\text{edge}} = 0.1.$$

The LightGBM model predicts logarithmic growth in in-strength, which is subsequently converted into a forecasted future in-strength. Predicted absolute and relative growth are then combined with historical network momentum, bibliometric indicators, and patent-derived information to produce two complementary rankings.

The objective is not to provide definitive predictions of future technological outcomes, but to identify research clusters whose current scientific dynamics, network evolution, and patent activity are consistent with future emergence.

4.2.1 Tier 1: Scientific Emergence Forecasting

Tier 1 identifies research areas that are already exhibiting sustained scientific momentum and are forecast to experience substantial future growth. The ranking combines predicted absolute and relative in-strength growth with recent historical growth, FWCI, and mean citation impact.

Table 4.8 presents the ten highest-ranked Tier 1 candidates forecast for 2028.

The Tier 1 ranking places research areas related to quantum communication, computation, cryptography, information processing, and hardware among the highest-scoring prospective candidates. Quantum communication receives the highest Tier 1 score because its current features and model-predicted growth produce the strongest combined emergence score.

Quantum computing receives one of the largest predicted increases, with in-strength forecast to rise from approximately 117.0 in 2025 to 803.2 in 2028. Quantum-information processing and variational quantum algorithms are also assigned substantial predicted growth by the model. These values represent

Table 4.8: Top-10 Tier 1 model generated scientific-emergence candidates forecast for 2028. $\widehat{k}_{2028}$ denotes forecasted in-strength, k_{2025} denotes current in-strength, and $\widehat{\Delta}$ denotes predicted absolute growth.

Rank	Research topic	$\widehat{k}_{2028}$	k_{2025}	$\widehat{\Delta}$
1	Quantum communication	570.2	141.3	428.9
2	Quantum computing	803.2	117.0	686.3
3	Quantum secret sharing	138.6	49.0	89.6
4	Quantum-information processing	630.4	145.2	485.1
5	Variational quantum algorithms	489.7	96.2	393.5
6	Quantum gates	275.8	68.2	207.5
7	QKD networks	230.3	79.6	150.7
8	Transmon qubits	129.8	57.3	72.5
9	Photonic quantum computing	134.3	52.3	82.0
10	Potential quantum machine learning	413.7	241.4	172.2

forecasted network trajectories and should not be interpreted as evidence that such growth will necessarily occur.

Quantum communication and QKD networks appear separately in the ranking, meaning that the model identifies potential future growth at both the broader communication level and the more specific network-infrastructure level. Quantum secret sharing provides another model-generated candidate related to secure quantum communication.

Hardware-oriented areas, including quantum gates, transmon qubits, and photonic quantum computing, also appear among the Tier 1 candidates. Their presence suggests that the model does not concentrate its prospective predictions exclusively on algorithmic or theoretical research, although their actual development remains to be evaluated using future observations.

The Tier 1 ranking should not be interpreted solely according to forecasted final size. The scoring procedure also incorporates predicted relative growth, historical momentum, FWCI, and citation impact. Consequently, smaller clusters such as quantum secret sharing, transmon qubits, and photonic quantum computing can rank highly despite having lower forecasted in-strength than broader fields such as quantum computing.

4.2.2 Tier 2: Patent-Informed Technology Transition

Tier 2 identifies research clusters that combine predicted academic growth with increasing patent-related activity. The ranking incorporates recent patent

boost, patent-boost velocity, and the growth of patent-enhanced inter-cluster connections as complementary indicators of possible technological translation.

Table 4.9 presents the ten highest-ranked Tier 2 candidates forecast for 2028.

Table 4.9: Top-10 Tier 2 model generated patent-informed technology-transition candidates forecast for 2028. b denotes recent patent boost, $\dot{b}$ denotes patent-boost velocity, and $\dot{E}_{\text{pat}}$ denotes patent-edge velocity.

Rank	Research topic	$\widehat{k}_{2028}$	b	$\dot{b}$	$\dot{E}_{\text{pat}}$
1	Quantum federated learning	111.9	4.857	0.029	132
2	Variational quantum algorithms	489.7	0.585	0.585	101
3	Spin qubits	165.5	0.283	0.235	7
4	Quantum-information processing	630.4	0.459	0.178	71
5	Quantum gates	275.8	0.893	0.694	83
6	Quantum systems	1215.7	0.924	0.924	370
7	Quantum devices	780.7	1.726	1.726	90
8	Quantum sensors	142.1	0.338	0.338	17
9	Transmon qubits	129.8	0.392	0.237	13
10	Applied quantum machine learning	64.6	0.086	0.034	6

Quantum federated learning receives the highest Tier 2 score. Although its forecasted in-strength remains moderate, it exhibits the strongest recent patent boost among the selected candidates and substantial growth in patent-enhanced inter-cluster connections. This combination suggests that industrial activity around privacy-preserving and distributed quantum machine-learning methods may be developing faster than the corresponding academic network.

Variational quantum algorithms, quantum-information processing, and quantum gates appear in both the scientific and patent-informed dimensions of the forecast. Their presence in both rankings indicates that these areas combine strong academic growth with increasing technological activity.

The Tier 2 ranking also highlights several hardware-oriented fields. Spin qubits, transmon qubits, quantum systems, quantum devices, and quantum sensors all exhibit positive patent momentum. Quantum systems show the largest forecasted in-strength in Tier 2, increasing to approximately 1215.7 by 2028, and also display the highest patent-edge velocity, with 370 additional patent-enhanced connections.

Quantum devices exhibit the strongest patent-boost velocity among the hardware-oriented candidates, while quantum gates combine a relatively high recent patent boost with substantial patent-edge growth. These signals suggest continued industrial interest in the components and control mechanisms

required for practical quantum-computing systems.

4.2.3 Qualitative Interpretation of the 2028 Forecast

The prospective analysis produces several groups of model-generated candidates. Tier 1 particularly emphasizes quantum communication, computation, information processing, algorithms, and selected hardware areas. Tier 2 additionally highlights quantum federated learning, quantum devices, quantum sensors, and other areas exhibiting patent-related signals.

These patterns should be interpreted as hypotheses about possible future research and technological trajectories. In contrast to the retrospective 2022–2025 experiment, no observed 2028 network is currently available against which these predictions can be validated. Consequently, the presence of a research area in Tables 4.8 and 4.9 indicates that it satisfies the emergence criteria of the proposed model; it does not demonstrate that the area will actually undergo the predicted development.

The two tiers provide complementary interpretations of these prospective candidates. Tier 1 emphasizes clusters whose scientific and network indicators are consistent with future scientific growth, whereas Tier 2 highlights candidates for which patent activity provides an additional technological signal. Future observations will be required to determine whether the predicted 2028 growth occurs and whether the patent-informed candidates subsequently exhibit stronger technological development.

Chapter 5

Limitations and Future Improvements

Although the proposed framework demonstrates promising performance in forecasting emerging technologies, several limitations should be acknowledged. Addressing these limitations presents opportunities for future improvements.

5.1 Limitations

Despite the concept-cleaning process, noise introduced during concept extraction remains a challenge. For example, the fragment "mml xmlns xlink ion" are still present in our concept dataset, even though it is clearly an extraction artifact rather than a meaningful technological concept, we can also find concept like "quantum system xmlns" which can be put into a cluster full of "xmlns" instead of the "quantum system" cluster.

This issue primarily stems from the manual construction of the stop-word list used during preprocessing. The cleaning process relied on manually identifying terms and patterns that should be removed from the concepts extracted by SciBERT. For instance, common verb fragments such as "sends", "processes", and "includes" were excluded because they frequently appear in abstracts without representing technological concepts. Similarly, leading abbreviation tokens such as "xss" and "sql" were removed when they appeared as isolated artifacts rather than meaningful concepts.

Nevertheless, manually identifying and eliminating every noisy concept is impractical, as it would require inspecting the entire concept vocabulary in-

dividually. Such a process would be time-consuming, difficult to scale, and potentially subjective. Consequently, a small number of extraction artifacts remain in the dataset and may occasionally appear among the highest-ranked forecasted concepts.

Similarly, another limitation concerns the presence of semantically overlapping clusters that are not merged and therefore remain as independent entities representing the same underlying concept. For example, cluster 7010, labeled "quantum key distribution", and cluster 4147, labeled "qkd protocol", describe essentially the same technological concept. However, these clusters were not combined during the clustering process. For example, we can find the cluster 7010 which achieved top 15 in tier 1 and top 17 in tier 2 but also find cluster 4147 which achieved top 22 in tier 2.

This issue may occur when the distance between cluster centroids exceeds the merging threshold, even though the clusters remain semantically related. In such cases, two highly similar concepts can be represented as separate clusters instead of a single consolidated technology domain. As a consequence, the associated scientific activity, growth, and forecasting indicators may be distributed across multiple clusters, potentially lowering the individual scores of each cluster and underestimating the importance of the broader concept. This is tied to the previous limitation where a concept can be named "Quantum key distribution xmlns" and thus be separated from the quantum key distribution cluster.

A second limitation is that the validation presented here remains limited to the quantum field. To improve the detection of emerging technologies, the methodology should be extended to a broader range of topics and scaled to multiple scientific domains. As discussed in the previous chapter, many papers cite works that are not included in the current dataset. This limitation may result in missing important information, particularly the connections between different research areas that could lead to the emergence of new technologies.

For example, if a quantum-related paper cites multiple works from the artificial intelligence domain, this relationship may indicate a growing interaction between these two fields. Such a connection could reveal an emerging area, such as quantum machine learning, which may then form a highly ranked cluster. However, expanding the analysis to multiple domains significantly increases the size of the dataset, the number of abstracts to process, and the complexity of the resulting concept clusters and citation networks. Consequently, the computational cost can become substantial.

Therefore, it is not feasible to include all scientific domains simultaneously, and careful decisions must be made regarding which domains should be selected for analysis. These choices should balance the objective of identifying cross-disciplinary emerging technologies with the computational constraints of the proposed methodology.

A similar limitation concerns the construction of the temporal citation graph. The current implementation updates the graph incrementally and therefore requires the complete graph structure to remain in memory while new yearly edges are added. Because the number of citation edges grows rapidly over time, this approach becomes memory intensive. To ensure that the graph could be constructed on the available hardware, two pruning strategies were introduced. First, edges with weights below a minimum threshold were discarded. Second, for each source cluster, only the top-ranked outgoing edges were retained. These pruning operations substantially reduce the size of the graph while preserving its strongest structural relationships. The experiments were conducted locally on a laptop equipped with 16 GB of RAM. Under these hardware constraints, the graph could contain approximately 4.5 million edges before memory exhaustion caused the program to terminate. Although the adopted pruning strategy enabled the complete forecasting pipeline to execute successfully, some weak citation relationships are inevitably removed. Future work could overcome this limitation by employing distributed graph processing, out-of-core graph storage, graph databases, or high-memory computing environments, allowing substantially larger scientific networks to be constructed without aggressive edge pruning.

Another limitation concerns the dependence of the proposed framework on several manually selected or empirically tuned parameters. The concept clustering, temporal stabilization, patent integration, and candidate selection stages rely on parameters such as clustering hyperparameters, similarity thresholds, smoothing coefficients, and filtering criteria. Although the main forecasting parameters were selected through validation experiments, an exhaustive sensitivity analysis was not conducted for every parameter of the pipeline. Consequently, some parameter choices may be specific to the quantum computing dataset and may require recalibration when applying the framework to other technological domains. Future work could therefore investigate systematic sensitivity analyses or data-driven parameter optimization to evaluate the robustness of the identified emerging technologies across different parameter configurations.

Another limitation identified in the proposed methodology concerns the α

coefficients used to weight the contribution of patents to both nodes and edges. The analysis shows that varying these coefficients has a limited impact on the final results, with the obtained rankings remaining highly similar and changing by less than 1% of the mean global growth percentile. This limited influence is mainly explained by the small proportion of links receiving a patent-based boost and by the relatively low magnitude of the applied boost.

Indeed, in most cases, only a limited number of patents contribute to boosting a given edge compared with the overall number of edges and scientific abstracts in the dataset. This issue is partly related to the differences in language and terminology between scientific papers and patents, which makes the alignment between both sources challenging. Additionally, the imbalance between the volume of scientific publications and patents further reduces the relative influence of patent information on the final scores.

As a result, patent information may not sufficiently impact the emergence score of certain technologies. For instance, an emerging technology may receive a patent-based boost, but if this contribution remains small and the initial score derived from scientific publications is also low, the final emergence score may remain insufficient for detection by the proposed methodology. Consequently, some technologies with strong industrial potential could be underestimated or overlooked. However, as shown in the previous chapter, patents do have an impact on the results, and the approach of using them as a boosting mechanism is effective. Nevertheless, this impact appears to operate primarily through their presence rather than their magnitude. Whether a cluster or an edge receives any patent-derived signal significantly influences the identification of emerging technologies, whereas, once this signal is present, fine-tuning its exact contribution through the α coefficients provides comparatively limited additional discriminative power. In other words, the binary presence or absence of patent information appears to be more important than the strength of the applied boost.

An additional limitation lies in the candidate selection step preceding the emergence scoring itself. Both Tier 1 and Tier 2 explicitly exclude clusters already situated in the top quartile of current size or connectivity (e.g., in-degree or pagerank), under the assumption that such clusters represent already-established fields rather than emerging ones. While this filtering helps avoid flagging dominant, mature technologies as "emerging," it also means that any cluster that is simultaneously large and still accelerating rapidly is structurally excluded from consideration, regardless of its growth trajectory. Future work

could revisit this fixed quartile-based exclusion, for instance by conditioning the filter on growth rate rather than absolute size alone, or by introducing a separate category for large, still-accelerating fields that does not rely on excluding them from the candidate pool entirely.

5.2 Future Improvements

In this study, SciBERT was used for extracting concepts from both scientific papers and patents. However, SciBERT was primarily trained on scientific publications and may therefore not fully capture the specific linguistic characteristics of patent documents. A model specifically trained on patent data, such as PatentBERT, could potentially provide more suitable embeddings for patent abstracts, as it is designed to better represent the terminology and writing style commonly used in patent literature.

Nevertheless, using different embedding models for scientific papers and patents introduces an additional challenge. Since each model would generate embeddings within its own latent representation space, the resulting vectors would belong to different domains, making direct comparison, aggregation, and joint analysis difficult. Addressing this issue would require an additional domain adaptation step to align the two embedding spaces and enable meaningful integration of concepts extracted from both sources.

Developing such an alignment strategy represents an important direction for future work. Several approaches could be investigated, including domain adaptation techniques, knowledge graph-based alignment, or large language model (LLM)-based methods to establish connections between the two semantic spaces. This could enable the use of PatentBERT for patent concept extraction while maintaining compatibility with scientific literature representations.

Another potential improvement concerns the integration of patent-based boosting mechanisms within the proposed methodology. In this work, we demonstrated that incorporating patent information influences the forecasting results; however, the specific formulation of the patent boost factor had a limited impact on the final outcomes. This suggests that the way patent influence is represented could be further improved. Rather than relying on a hand-set, fixed boosting formula, future work could explore adaptive or learned boosting methods, where the contribution of patent signals is optimized directly from data rather than manually specified. In parallel, instead of relying primarily on

concept co-occurrence relationships, future approaches could explore richer patent networks, such as citation graphs. Patent citation networks can capture technological lineage by connecting patents to prior patents as well as scientific publications, potentially providing a more meaningful representation of technological development and influence.

Another natural extension of this work would be to perform forecasting at the concept level within the highest-ranked clusters. The current methodology identifies emerging research clusters, which often correspond to broad topics such as *Quantum communication* or *Quantum computing*. While these clusters effectively capture major technological directions, they do not reveal which specific concepts within each cluster are driving the predicted growth.

A second-stage ranking could therefore be applied to the concepts belonging to the most promising clusters by leveraging concept-level citation dynamics, embedding representations, and temporal features. This finer-grained analysis would make it possible to identify niche concepts that are likely to emerge before they become dominant research themes. Such an approach could provide more actionable insights for researchers, funding agencies, and industry by highlighting specific techniques, algorithms, or applications that are expected to gain importance within a broader emerging research area.

Finally, incorporating additional data sources could provide complementary information that is not captured in scientific abstracts or patent documents. For instance, job postings can reveal the existence of market demand for specific expertise, indicating that a technological domain requires particular skills and that related concepts may be emerging. Similarly, information about companies associated with a given technology, such as funding activity, investment trends, or the increasing number of companies operating in a specific field, can provide additional signals of technological growth and maturity.

Integrating company-level data from sources such as Crunchbase could therefore improve the richness of the dataset while also providing an external validation mechanism for the identified emerging technologies. However, incorporating these heterogeneous data sources would require an additional processing layer involving data collection, parsing, cleaning, and analysis. Furthermore, combining information from different domains would introduce new challenges related to data consistency, temporal alignment, and the integration of heterogeneous indicators into the existing forecasting framework.

Chapter 6

Conclusion

This study demonstrates the potential of integrating patent data with academic literature to forecast emerging and disruptive technologies. By combining papers and patents within a shared technological domain, a patent-derived boosting mechanism was introduced and applied to both nodes and edges of a citation graph constructed from academic references. SciBERT was used to extract key concepts from paper abstracts, while a clustering and stabilization pipeline was developed to reduce the noise associated with embedding-based concept extraction and to ensure consistent concept labels across successive years. These stable concept clusters formed the basis of a temporal citation graph, on which a LightGBM forecasting model was trained using walk-forward cross-validation.

Applied to the domain of quantum computing, the proposed pipeline produced encouraging results. The two-tier detection framework—consisting of a model-driven Rising Fields tier and a Patent-Informed Technology Transition tier successfully identified research clusters that experienced substantial growth during the 2022–2025 validation period. Notably, the ablation analysis revealed that citation-network and bibliometric signals alone were already sufficient to identify many high-growth candidates; the primary contribution of patent information lay instead in sharpening the ranking of these candidates, nearly doubling the rank correlation with realized future growth. This suggests that the value of patent data in technology forecasting is best understood as a refinement signal rather than a strict precondition for detection.

Despite these promising findings, several limitations should be acknowledged. First, the methodology currently relies exclusively on OpenAlex as its source of academic publications. Although OpenAlex provides broad coverage, its indexing may introduce systematic biases toward particular disciplines, pub-

lication venues, or geographic regions. Second, a semantic gap remains between patents and academic publications. Patent documents are written primarily for legal protection rather than scientific communication, resulting in vocabulary and writing styles that differ substantially from those of research papers. This discrepancy limits the effectiveness of direct embedding-based alignment between the two corpora. Third, the proposed patent integration strategy applies patent signals as multiplicative weights, representing a deliberate yet simplified design choice. Future work could instead learn the contribution of patent information directly as part of the forecasting model. Finally, the construction of the concept stop list remains partially manual and domain-specific, limiting the degree of automation when transferring the pipeline to other research areas.

Nevertheless, the proposed framework is largely domain-agnostic in its overall architecture. Applying it to another scientific field primarily requires updating the initial OpenAlex topic identifiers and adapting domain-specific preprocessing. However, some clustering, similarity, and filtering parameters may require recalibration to account for differences in the structure and terminology of other technological domains.

Several avenues for future research emerge from this work. Additional sources of innovation-related information, such as company jobs posting, company funding information, industrial reports could be incorporated to further enrich and validate technology forecasts. The semantic alignment between patents and scientific publications could be improved through transformer models specifically trained on patent corpora, such as PatentBERT, enabling more accurate cross-domain concept matching. Finally, more expressive approaches to patent integration could be explored by representing patent relationships as an additional graph modality and learning their interaction with the academic citation graph end-to-end, rather than incorporating patent information through a handcrafted boosting mechanism.

Bibliography

[1] J. Bellasio and E. Silfversten, *The Impact of New and Emerging Technologies on the Cyber Threat Landscape and Their Implications for NATO*, RAND Corporation, 2020.

[2] H. Dylan and N. Stivang, "Emerging technologies and national security intelligence," *Intelligence and National Security*, vol. 40, no. 6, pp. 988–1009, 2025.

[3] X. Gu and M. Krenn, "Interesting scientific idea generation using knowledge graphs and LLMs: Evaluations with 100 research group leaders," *arXiv preprint arXiv:2405.17044*, 2024.

[4] T. Maillart, T. Chataing, D. Dosu, P. Bagourd, J. Jang-Jaccard, and A. Mermoud, "Forecasting conceptual diffusion in science: The case of quantum computing," *arXiv preprint arXiv:2606.03919*, 2026.

[5] Z. Yang *et al.*, "A deep learning-based method for predicting the emerging degree of research topics using emerging index," *Scientometrics*, vol. 129, pp. 4021–4042, 2024.

[6] L. Dolamic, J. Jang-Jaccard, A. Mermoud, and V. Lenders, "Automated identification of emerging technologies: Open data approach," in *Joint Workshop of EEKE 2024 and AI+Informetrics*, 2024.

[7] E. Norouzi, S. Hertling, and H. Sack, "Conexion: Concept extraction with large language models," *arXiv preprint arXiv:2504.12915*, 2025.

[8] T. Daim, H. Yalcin, and A. Mermoud, "Monitoring cybersecurity technology through the years: A technology mining approach through bibliometrics and patent analysis," *Journal of Cyber Security Technology*, 2024.

[9] OpenAlex, "OpenAlex," Available: `https://openalex.org`

[10] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[11] S. Rose, D. Engel, N. Cramer, and W. Cowley, "Automatic keyword extraction

from individual documents," in *Text Mining*, M. W. Berry and J. Kogan, Eds. John Wiley & Sons, 2010.

[12] J. Ramos, "Using TF-IDF to determine word relevance in document queries," in *Proc. First Instructional Conference on Machine Learning*, pp. 29–48, 2003.

[13] R. Campos, V. Mangaravite, A. Pasquali, A. Jorge, C. Nunes, and A. Jatowt, "YAKE! Keyword extraction from single documents using multiple local features," *Information Sciences*, vol. 509, pp. 257–289, 2020.

[14] N. Giarelis, N. Kanakaris, and N. Karacapilidis, "A comparative assessment of state-of-the-art methods for multilingual unsupervised keyphrase extraction," in *Proc. 17th IFIP Int. Conf. Artificial Intelligence Applications and Innovations (AIAI)*, pp. 635–645, 2021.

[15] R. Mihalcea and P. Tarau, "TextRank: Bringing order into text," in *Proc. EMNLP*, 2004, pp. 404–411.

[16] X. Wan and J. Xiao, "CollabRank: Towards a collaborative approach to single-document keyphrase extraction," in *Proc. COLING*, 2008, pp. 969–976.

[17] M. Grootendorst, *KeyBERT: Minimal Keyword Extraction with BERT*, Zenodo, 2020. doi:10.5281/zenodo.4461265.

[18] I. Beltagy, K. Lo, and A. Cohan, "SciBERT: A pretrained language model for scientific text," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3615–3620.

[19] J.-S. Lee and J. Hsiang, "Patent classification by fine-tuning BERT language model," *World Patent Information*, vol. 61, Art. no. 101965, 2020.

[20] A. Cohan, S. Feldman, I. Beltagy, D. Downey, and D. S. Weld, "SPECTER: Document-level representation learning using citation-informed transformers," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 2270–2282.

[21] L. Dolamic, J. Jang-Jaccard, A. Mermoud, and V. Lenders, "Automated identification of emerging technologies: OpenData approach," 2023.

[22] S. Xu, L. Hao, G. Yang, K. Lu, and X. An, "A topic models based framework for detecting and forecasting emerging technologies," *Technological Forecasting and Social Change*, vol. 162, Art. no. 120366, 2021.

[23] X. Gu and M. Krenn, "Forecasting high-impact research topics via machine learning on evolving knowledge graphs," *Machine Learning: Science and Technology*, vol. 6, no. 2, Art. no. 025041, 2025.

[24] M. Krenn, L. Buffoni, B. Coutinho *et al.*, "Forecasting the future of artificial intelligence with machine learning-based link prediction in an exponentially

growing knowledge network," *Nature Machine Intelligence*, vol. 5, pp. 1326–1335, 2023.

[25] Y. Zhou, F. Dong, Y. Liu *et al.*, "Forecasting emerging technologies using data augmentation and deep learning," *Scientometrics*, vol. 123, pp. 1–29, 2020.

[26] J. Shlens, "A tutorial on principal component analysis," *arXiv preprint arXiv:1404.1100*, 2014.

[27] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform manifold approximation and projection for dimension reduction," *arXiv preprint arXiv:1802.03426*, 2018.

[28] R. J. G. B. Campello, D. Moulavi, and J. Sander, "Density-based clustering based on hierarchical density estimates," in *Advances in Knowledge Discovery and Data Mining (PAKDD 2013)*, J. Pei, V. S. Tseng, L. Cao, H. Motoda, and G. Xu, Eds., *Lecture Notes in Computer Science*, vol. 7819, Springer, Berlin, Heidelberg, 2013, pp. 160–172.

[29] C. C. Holt, "Forecasting seasonals and trends by exponentially weighted moving averages," *International Journal of Forecasting*, vol. 20, no. 1, pp. 5–10, 2004.

[30] S. Brin and L. Page, "The anatomy of a large-scale hypertextual Web search engine," *Computer Networks and ISDN Systems*, vol. 30, nos. 1–7, pp. 107–117, 1998.

[31] T. U. Daim, G. Rueda, H. Martin, and P. Gerdsri, "Forecasting emerging technologies: Use of bibliometrics and patent analysis," *Technological Forecasting and Social Change*, vol. 73, no. 8, pp. 981–1012, 2006.

[32] B. Yoon and Y. Park, "A systematic approach for identifying technology opportunities: Keyword-based morphology analysis," *Technological Forecasting and Social Change*, vol. 72, no. 2, pp. 145–160, 2005.

[33] A. Ebadi, A. Auger, and Y. Gauthier, "Detecting emerging technologies and their evolution using deep learning and weak signal analysis," *Journal of Informetrics*, vol. 16, no. 4, Art. no. 101344, 2022.

[34] A. Ghaemmaghami, A. Schiffauerova, and A. Ebadi, "Detecting emerging technologies in artificial intelligence scientific ecosystem using an indicator-based model," *arXiv preprint arXiv:2211.01348*, 2022.

[35] F. Narin, K. S. Hamilton, and D. Olivastro, "The increasing linkage between U.S. technology and public science," *Research Policy*, vol. 26, no. 3, pp. 317–330, 1997.

[36] M. Meyer, "Patent citations in a novel field of technology—What can they

tell about interactions between emerging communities of science and technology?" *Scientometrics*, vol. 48, no. 2, pp. 151–178, 2000.

[37] J. A. Hartigan and M. A. Wong, "Algorithm AS 136: A K-Means clustering algorithm," *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, vol. 28, no. 1, pp. 100–108, 1979.

[38] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD)*, 1996, pp. 226–231.

[39] M. Grootendorst, "BERTopic: Neural topic modeling with a class-based TF-IDF procedure," *arXiv preprint arXiv:2203.05794*, 2022.

[40] R. Egger and J. Yu, "A topic modeling comparison between LDA, NMF, Top2Vec, and BERTopic to demystify Twitter posts," *Frontiers in Sociology*, vol. 7, Art. no. 886498, 2022.

[41] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent Dirichlet allocation," *Journal of Machine Learning Research*, vol. 3, pp. 993–1022, 2003.

[42] D. Angelov, "Top2Vec: Distributed representations of topics," *arXiv preprint arXiv:2008.09470*, 2020.

[43] H. Geng, D. Wang, F. Zhuang, X. Ming, C. Du, T. Jiang, *et al.*, "Modeling dynamic heterogeneous graph and node importance for future citation prediction," in *Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM)*, 2022, pp. 572–581.

[44] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.